
PHYS 475

Cosmology

Prof. Ashley Blemis

figures/notescovers.pdf

Typeset by Aiden Diaz

Credit to Prof. Ashley Blemis for His Lectures on PHYS 475

Preface

A majority of the information presented in this note package is transcribed directly from the *PHYS 475: Cosmology* lectures during the Spring 2025 term, taught by Prof. Ashley Blemis.

These notes are by no means exhaustive nor are they guaranteed to be error-free, and I do not make any claims about the reliability or accuracy of these set of notes. They are intended primarily for personal use and for those who may benefit from an alternative presentation of the material.

Please use these notes at your own discretion.

Contents

1	The Fundamental Observations	1
1.1	Olber's Paradox.	1
1.2	The Cosmological Principle.	2
1.3	The Expanding Universe.	2
1.4	Types of Particles	3
1.5	The Cosmic Microwave Background (CMB)	4
2	Newtonian to Einsteinian View	5
2.1	The Short-Range Forces	5
2.2	The Long-Range Forces	5
2.3	The Newtonian Field Equations	5
2.4	Special Relativity and Spacetime	6
3	The Geometry of Spacetime	8
3.1	Curvature	8
3.2	The Robertson-Walker Metric	11
3.3	Redshift, the Scale Factor, and Friedmann	13
3.4	The Fluid and Acceleration Equations	16
3.5	Equation of State	17
3.6	Energy Evolution and the Cosmological Constant	18
3.6.1	Dark Energy Dominated Universe	20
3.7	Model Universes	21
3.7.1	Model 1: The Empty Universe.	21
3.7.2	Model 2: Flat, Single-Component Universes	23
3.7.3	Model 3: Multi-Component Universes	25
3.7.4	Model 4: The Matter-Lambda Modern Model	27
3.7.5	The Benchmark Model	30
4	Cosmological Parameters	32
4.1	Kinematics of the Scale Factor	32
4.2	Variable Measurement	34
4.2.1	Measuring H_0	34
4.2.2	Measuring q_0	35
4.2.3	Independent Confirmation	36
4.3	The Missing Mass Problem	36

4.4	Gravitational Lensing	41
5	The Cosmic Microwave Background	42
6	Big Bang Nucleosynthesis (BBN)	44

1 The Fundamental Observations

1.1 Olber's Paradox.

"If the universe is infinite, eternal, and uniformly filled with stars, why is the entire night sky not as bright as the surface of the sun?"

To rectify Olber's paradox, we will rely on a couple definitions. Flux from a star,

$$f = \frac{L_*}{2\pi r^2}, \quad (1)$$

angular area of a star (in steradian),

$$\Omega = \frac{\pi R_*^2}{4\pi r^2}, \quad (2)$$

and the surface brightness of a star,

$$\Sigma_* = \frac{f}{\Omega} = \frac{L_*}{\pi R_*^2}. \quad (3)$$

We must now look at four assumptions which we have made to construct Olber's Paradox.

Absorption. We failed to account for the fact that anything that absorbs starlight will inevitably heat up until it is as hot as the stars itself. This, in turn, leads to heat emission and therefore light being emitted from the body itself.

Finite Size. We assume a finite volume, and so we thus have a limited number of stars. Many lines of sight will not end on a star, which may be a solution to our incorrect assumption.

Finite Age. If the universe began a finite time ago, light from extremely distance stars has not yet had time to reach us. Thus, we only see a horizon distance.

Suppose now we are interested in the expected distance to a star in our universe. Why is this important? If our universe is uniformly filled with stars and *not* infinitely bright, we can expect our line of sight to, on average, land on a star at some specific expectation value. To do this, we will define two important properties.

Let us assume that the stellar number density of stars is ρ_* ¹ and that each star is a sphere with a spherical radius of R_* . Then, the probability of our line of sight hitting a star in a shell of thickness dr is

$$dP = \frac{\text{size of star}}{\text{size of shell}} (\text{number of shells}) = (\pi R_*^2 \rho_*) dr = \frac{dr}{l}$$

where the general definition of the mean free path is

$$l \equiv \frac{1}{\pi \sigma} = \frac{1}{\pi R_*^2 \rho}.$$

¹Note that $[\rho_*] = [*/[V]]$ where V is volume.

If we consider a large number of lines of sight, I , pointing radially outwards and travelling through space, we expect some of them to be blocked by stars. The change in the number of these unblocked lines is dI , and through a shell of thickness dr , we expect the relation

$$dI(r) = -I(r)(\rho_* \pi R_*^2) dr = -\frac{I(r)}{l} dr$$

where we attach a negative sign for decreasing number of unblocked lines. We thus arrive at the differential equation

$$\frac{dI(r)}{dr} = -\frac{I(r)}{l}$$

which has the solution

$$I(r) = I_0 e^{-r/l}.$$

Using this, the probability distribution function we are interested in (probability per unit length), $F(r)$, in some interval $[r, r + dr]$ is thus given by

$$F(r) dr = \frac{e^{-r/l}}{l} dr.$$

Then, the mean distance to a star along any line of sight is simply

$$\langle r \rangle = \int_0^\infty r F(r) dr = \int_0^\infty \frac{r}{l} e^{-r/l} dr.$$

By IBP, we have

$$\langle r \rangle = \frac{1}{l} \left[-r l e^{-r/l} \Big|_0^\infty + \int_0^\infty l e^{-r/l} dr \right] = \frac{1}{l} \left[-l^2 e^{-r/l} \Big|_0^\infty \right] = l$$

We thus conclude that in a static, infinite universe, every line of sight would end on a star, and in fact,

$$\langle r \rangle = l = 10^{18} \text{ Mpc}.$$

1.2 The Cosmological Principle.

We must define the following qualities of our universe. Firstly, the universe is **isotropic**. That is, it looks the same in all directions. The universe is also **homogeneous**, since it looks the same in all directions (but is only true on scales larger than 100 Mpc). Lastly, the universe follows the **Copernican Principle**. We are not able to occupy a special or privileged place within the universe itself.

Combining these qualities, we can make the conclusion that our universe is homogeneous.

1.3 The Expanding Universe.

We say that the **redshift** in z of light from distant objects is given by

$$z \equiv \frac{\lambda_{\text{obs.}} - \lambda_{\text{emit.}}}{\lambda_{\text{emit.}}} \quad (4)$$

The governing equation for redshift is given by **Hubble's Law**, which says that a galaxy's recession velocity is linearly proportional to its distance, with an associated **Hubble Constant**,

$$H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This constant defines the uniformly expanding universe, which has units of inverse time. That is,

$$t = \frac{1}{H_0}. \quad (5)$$

1.4 Types of Particles

There are *six* important types of particles we observe in the universe.

Baryons and Leptons. **Protons** and **Neutrons** are **Baryon** particles (made of three quarks), while **electrons** are made of **Leptons**. In cosmology, we say that **Baryonic Matter** refers to all ordinary matter (p, n, or e). Under *neutron decay*, we find that free neutrons are unstable, such that

$$n \rightarrow p + e^-$$

with a decay time of about 15 minutes, $\tau_n \approx 880 \text{ s}$. Neutrons only survive inside bound atomic nuclei, and the universe has a **Cosmic Baryon Budget** of about 75% Hydrogen (unbound protons) and 25% Helium.

Neutrinos. There are *three* types of neutrinos: Electron Neutrinos (ν_e), Muon Neutrinos (ν_μ), and Tau Neutrinos (ν_τ). Neutrinos interact very weakly with other matter, where each type of neutrino is actually a quantum superposition of three different mass states (ν_1, ν_2, ν_3).

Uniquely, neutrinos can transform into other types of themselves, which is called an **neutrino oscillation**. This is only possible if the neutrino has mass.

The neutrino mass is generally small but nonzero, with a condition and upper limit,

$$0.057 \text{ eV} \leq \sum m_\nu c^2 \leq 0.3 \text{ eV}$$

which allows us to conclude that they are not simply enough to be a primary component of dark matter.

Photons. Photons are massless particles of light, governed by the energy relation

$$E_\gamma = hf.$$

In a system within **thermal equilibrium**, photons have well-defined temperature and a **blackbody spectrum** governed by Planck's Law,

$$\epsilon(f) df = \frac{8\pi h}{c^3} \frac{f^3}{\exp(hf/kT) - 1} df. \quad (6)$$

Energy density in a frequency range of df is given by

$$\epsilon_\gamma = \alpha T^4$$

and gives the Stefan-Boltzmann law after integrating over all frequencies.

1.5 The Cosmic Microwave Background (CMB)

The CMB is an isotropic background of microwave radiation, fitting a near-perfect blackbody spectrum. It is the relic of the **Big Bang**. There are four key measured properties of the CMB:

1. Temperature: $T_0 = 2.7255 \pm 0.0006$ K.
2. Energy Density: $\epsilon_r = 4.165 \cdot 10^{-14}$ Jm⁻³.
3. Number Density: $n_\gamma = 4.107 \cdot 10^8$ m⁻³.
4. Mean Energy: $E_{\text{mean}} = 6.344 \cdot 10^{-4}$ eV.

We can derive how the CMB cools by the use of thermodynamics. The first law states

$$dQ = dE + P dV = 0 \quad (7)$$

where dQ is the heat added to our system, dE is the change in total internal energy of the system, and $P dV$ is the work done by the system as it expands or contracts. We assume that the expansion is **adiabatic** since the universe is homogeneous, so we can simply set $dQ = 0$.

Given that our proton energy is $E = \epsilon_\gamma V = \alpha T^4 V$ and our photon pressure is $P = \epsilon_\gamma/3 = \alpha T^4/3$. Plugging these into our equation and using the fact that $V \propto a(t)^3$, where $a(t)$ is the time-dependent scale factor of our universe (how the universe increases linearly in size). We first calculate dE/dt by product rule since $T = T(t)$ and $V = V(t)$,

$$\frac{dE}{dt} = \frac{dE}{dV} \frac{dV}{dt} + \frac{dE}{dT} \frac{dT}{dt} = \alpha T^4 \frac{dV}{dt} + 4\alpha T^3 V \frac{dT}{dt}$$

and since $dE/dt = -P(dV/dt)$, we have

$$\begin{aligned} \alpha T^4 \frac{dV}{dt} + 4\alpha T^3 V \frac{dT}{dt} &= -\frac{\alpha T^4}{3} \frac{dV}{dt} \\ T^4 \frac{dV}{dt} + 4T^3 V \frac{dT}{dt} &= -\frac{T^4}{3} \frac{dV}{dt} \\ 4T^4 V \frac{dT}{dt} &= -\frac{4}{3} T^4 \frac{dV}{dt} \end{aligned}$$

and so we arrive at

$$\frac{1}{T} \frac{dT}{dt} = -\frac{1}{3V} \frac{dV}{dt}.$$

Using the proportionality $V \propto a(t)^3$, we can say $\ln(V) \propto 3 \ln[a(t)]$ and so

$$\frac{d}{dt}[\ln(V)] \propto 3 \frac{d}{dt}[\ln(a)] \implies \frac{1}{V} \frac{dV}{dt} \propto \frac{3}{a} \frac{da}{dt}$$

to give

$$\frac{1}{T} \frac{dT}{dt} \propto -\frac{1}{a} \frac{da}{dt}.$$

Finally, we can solve this differential equation easily to give

$$\boxed{T \propto a(t)^{-1}}. \quad (8)$$

Another important quality of the CMB is its transition to opaqueness to transparency. At about $T \approx 3000$ K, protons and electrons combined to form neutral hydrogen; this, in turn, caused the universe to become **transparent** since there were few free electrons to scatter light.

Blackbody photons were released as a result, and continue to do so until today. However, their temperature remains red shifted by a factor of about $z = 1090$ to $T \approx 2.7$ K.

2 Newtonian to Einsteinian View

We know that there are two views of gravity thus far: the **Newtonian** view, and the **Einsteinian** view. The former views gravity as a *force*, acting between massive objects in a fixed, absolute space and time. Furthermore, $F \propto a$. The latter views gravity as the *curvature of spacetime*. Space and time are dynamic and interwoven, and spacetime itself tells *mass-energy* how to move.

2.1 The Short-Range Forces

The Strong Nuclear Force The strong nuclear force binds quarks into protons and neutrons, and binds nuclei overall together. It has a range of $\approx 10^{-15}$ m, so we effectively cannot see it at large scales. However, it was extremely important during the **Quark Epoch**, which occurred in the universe in the range $t < 10^{-6}$ s and the **Big Bang Nucleosynthesis** in the range $t \approx 3$ min.

The Weak Nuclear Force The weak nuclear force governs radioactive decay and nuclear reactions (e.g. $n \rightarrow p + e^- + \bar{\nu}_e$), however it has a even shorter range at $\approx 10^{-18}$ m. The weak nuclear force played a large role in the universal age $t < 1$ s.

2.2 The Long-Range Forces

The Electromagnetic Force The EM force is extremely powerful and infinite in its range. However, on large scales, the universe is effectively electrically neutral, as the attractive and repulsive forces cancel perfectly. The EM kept the universe in an opaque plasma before its **Recombination** period, at $t < 370,000$ yr.

The Gravitational Force Like the EM force, the gravitational force is infinite in range. It is also **always** attractive. In this way, it is a universal force and cumulative in nature, adding up in larger scales. Gravity has played a major role at the end of the **Inflationary Epoch** up to the present day, from $t \approx 10^{-34}$ s to $t = t_0$, where t_0 is the present time.

2.3 The Newtonian Field Equations

In the Newtonian view, we can represent the gravitational potential of an object by Poisson's equation, in the form

$$\nabla^2 \phi = 4\pi G \rho \quad (9)$$

where ρ is the mass density of the object and ϕ is the gravitational potential field. We can solve this in its integral form, which would be

$$\phi(\mathbf{r}) = -G \int \frac{\rho(\mathbf{x})}{|\mathbf{x} - \mathbf{r}|} d^3\mathbf{x}. \quad (10)$$

Furthermore, we know that $\mathbf{F} \propto \mathbf{a}$, but at the same time $\mathbf{F}$ is the gradient of potential, so we also have

$$\mathbf{a} = -\vec{\nabla} \phi(\mathbf{r}). \quad (11)$$

Notice the negative, which indicates that objects accelerate in the direction of steepest potential descent.

2.4 Special Relativity and Spacetime

Recall the relativistic transformations that keep the speed of light, c , constant for all observers,

$$\begin{aligned} x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\left(t - \frac{vx}{c^2}\right) \end{aligned} \tag{12}$$

where $\gamma = 1/\sqrt{1 - v^2/c^2}$. The formulation of special relativity (SR) consequently tells all observers that the measurements of time and space are different, relative to each other. We call this *time dilation* and *length contraction*. On the other hand, **spacetime separation** is the same between any two events.

We can represent this spacetime separation in 4D spacetime by

$$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \tag{13}$$

Here, the minus sign is extremely important and is what defines the geometry of spacetime, which we all **Minkowski Space**.

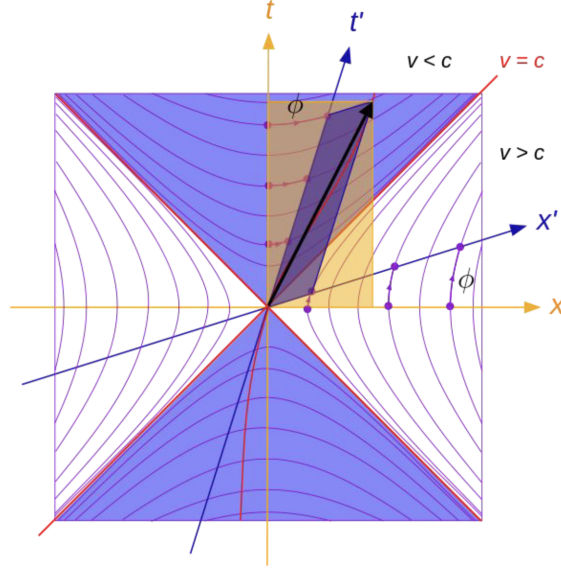


Figure 1: Geometric representation of spacetime.

To develop general relativity (GR), we will need three physical principles:

1. **Mach's Principle:** an object's inertia is determined by all the matter in the universe.
2. **General Covariance:** the laws of physics must be the same for **all** observers, even those who are acceleration.
3. **The Einsteinian Equivalence Principle.**

In particular, the third principle states that an observer in an inertial reference frame cannot distinguish between the acceleration of the system or the acceleration of the body itself. The implication for light, is that gravity must bend the path of light itself in both situations.

Curve spacetime tends to affect this by causing light to travel in **geodesics** (in particular, the shortest paths). In a flat, Euclidean space (which we are not), a geodesic is a straight line. In 3D space as in reality, light travels in a curve, to which we can conclude the space is non-Euclidean. That is, mass-energy causes spacetime to curve.

3 The Geometry of Spacetime

The framework of General Relativity (which is vital to the study of cosmology, though not used much in this course), is determined by the mass-energy of the universe. In particular, it necessitates the use of the **metric** which tells us the distance between two nearby points in space.

3.1 Curvature

The Cosmological Principle allows us to consider spaces of *uniform curvature*, which we denote with κ . For a 2D surface, there are only three possibilities: flat surface ($\kappa = 0$), positively curved surface ($\kappa = 1$), and a negatively curved surface ($\kappa = -1$).

Flat Space. The case of $\kappa = 0$ is simple. We can imagine a flat plane whose geodesic is a straight line. Of course, the space is Euclidean and so it follows that the internal angles of a triangle sum to π . That is,

$$\alpha + \beta + \gamma = \pi \quad (14)$$

for some angles α , β and γ . Here, the metric is simple and is given by the Pythagorean Theorem, which states

$$(dl)^2 = (dr)^2 + r^2(d\theta)^2. \quad (15)$$

We can derive this identity explicitly by considering the magnitude of a vector, dl , with components given by dr and $r d\theta$.

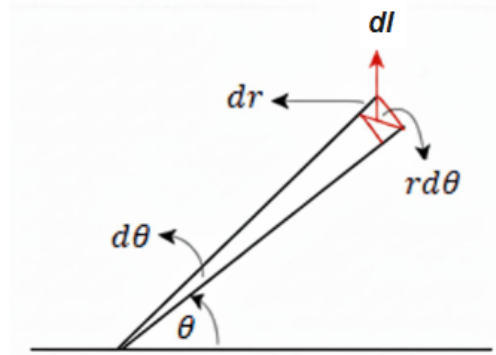


Figure 2: Deriving the flat space metric.

Here, dr is the radial component of the vector dl and $r d\theta$ is the tangential component. Of course, since the space is Euclidean, all components are perpendicular and we thus arrive at the metric before.

Positively Curved Space. Consider now the case $\kappa = 1$. Here, we can imagine the surface of a sphere whose geodesic the arc of a great circle.² Now, our space is non-Euclidean, and so the internal angles of a triangle within the sphere is given by

$$\alpha + \beta + \gamma = \pi + \frac{A}{R^2} \quad (16)$$

²Be careful to note that this is **not** 3D. The "surface" lies in 3D, but the surface itself is 2D, which we are interested in.

where A is the area of the triangle and R is the radius of the sphere. We can use again the Pythagorean theorem (it still works, but will give a weird result) on our surface. The difference here is that the tangential step will be much shorter. In fact, the tangential step is affected by a geometric factor given by $\sin(r/R)$.

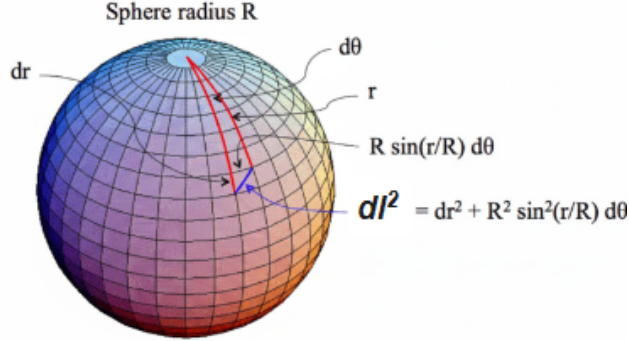


Figure 3: Deriving the positively curved space metric.

That is, the Pythagorean theorem should give us

$$(dl)^2 + (dr)^2 + R^2 \sin^2\left(\frac{r}{R}\right) (d\theta)^2 \quad (17)$$

where the $\sin(r/R)$ factor is a mathematical signature of positive curvature.

Negatively Curved Space. Now consider the case $\kappa = -1$. Here, the surface is a hyperboloid or a "saddle", with a geodesic resembling a diverging curve. Similar to the positively curved case, the internal angles of a triangle on our surface sum to

$$\alpha + \beta + \gamma = \pi - \frac{A}{R^2} \quad (18)$$

where the parameters are the same as before. We can repeat a similar derivation for the metric, but the result simply depends only on $\sinh$ instead of $\sin$, and so we have

$$(dl)^2 = (dr)^2 + R^2 \sinh^2\left(\frac{r}{R}\right) (d\theta)^2. \quad (19)$$

Consider now what a hyperbolic space would affect what we observe. We are interested in the angular size, $d\theta$, of an object of length dl situated some distance away of r on a negatively curved surface. Beginning with our metric, we have

$$dl = S_\kappa(r) d\theta = R \sinh\left(\frac{r}{R}\right) d\theta.$$

Solving for the angular size, we have

$$d\theta = \frac{dl}{R \sinh(r/R)}.$$

Notice that if we take the derivative with respect to r ,

$$\frac{d\theta}{dr} = \frac{d}{dr} \left[\frac{dl}{R \sinh(r/R)} \right] = -\frac{\cosh(r/R)}{R^2 \sinh^2(r/R)} dl < 0$$

which implies that the angular size decreases as the distance r increases. At a large distance $r \gg R$, we can use a Taylor expansion to approximation $\sinh(x) \approx e^{-x}/2$, so that

$$d\theta \approx \frac{2 dl}{R e^{r/R}} = \frac{2}{R} e^{-r/R} dl \implies d\theta \propto e^{-r/R}.$$

So, we expect the apparent size of an object to shrink exponentially.

A major question is how we translate these 2D models to what we actually observe. There are two perspectives: the **bird's eye view** which 3D models space itself, or the **observer's view** which models the 2D celestial sphere we see from our single point.

We can connect these 2D analogies to cosmology by noting that in hyperbolic space, geodesics curve *outward*. We expect the angular size to shrink, so objects appear *smaller* than in flat space. In positively curved space, geodesics curve *inward*, we expect the angular size to increase, so objects will appear *larger* than in flat space. Of course, the flat space case is trivial.

We can generalize these 2D cases to 3D space. In fact, we have the same three possibilities. Flat space corresponds to $\kappa = 0$, while negatively curved space corresponds $\kappa = -1$, where R is the characteristic scale where geometry deviates from the flat space. On the other hand, positively curved space, $\kappa = 1$, corresponds to the space where a 3D surface comes from a 4D hypersphere of radius, R .

We can write our results in a compact form, where the metric is generally given by

$$(dl)^2 = (dr)^2 + S_\kappa(r)^2 (d\Omega)^2, \quad (d\Omega)^2 = (d\theta)^2 + \sin^2(\theta)(d\phi)^2 \quad (20)$$

and the function $S_\kappa(r)$ is such that

$$S_\kappa(r) = \begin{cases} R \sin(r/R), & \kappa = 1 \\ r, & \kappa = 0 \\ R \sinh(r/R), & \kappa = -1. \end{cases} \quad (21)$$

There are two ways to define the radius in curved space. What we have been using is the **circumferential radius**, $\hat{r}$, which is "inferred". This is the radius you would calculate from the circumference of a circle at that point, with the flat-space formula $C = 2\pi\hat{r}$. On the other hand, we have the **geodesic radius**, r , which is the "true" distance, the shortest path you would walk along the surface from an origin to a point. In curved space, there are **not** the same.

In the metric for 3D space, the function $S_\kappa(r)$ is the circumferential radius. In fact, $\hat{r} = S_\kappa(r)$. We can use this to understand how $S_\kappa(r)$ interacts with our universe. For flat space, we have $S_0(r) = r$, so the two radii are identical ($\hat{r} = r$). For positively curved space, we have $S_1(r) = R \sin(r/R)$, so $\hat{r} < r$ since the circumferential radius is **smaller** than the geodesic distance, and in hyperbolic space, we have $S_{-1}(r) = R \sinh(r/R)$, so $\hat{r} > r$ since the circumferential radius is **larger** than the geodesic distance.

We can consider an example of comparing the surface area of a sphere in a flat vs. negatively curved space to see how brightness changes. The surface area of a sphere (in circumferential radius) is given by $A(r) = 4\pi S_\kappa(r)^2$. We are interested in hyperbolic space, so we let $A(r) = 4\pi[R \sinh(r/R)]^2$. Thus, we find for $R = 100$ Mpc and $r = 50$ Mpc, that $A \approx 34200$ Mpc². Flat space nets us $A_{\text{flat}} = 4\pi r^2 \approx 31400$ Mpc². We can thus conclude that

$$\frac{A}{A_{\text{flat}}} \approx 0.09$$

so that hyperbolic space is about 9% larger. We expect the light to be more spread out, so a light source within this space would be dimmer.

However, we read a paradox: why does an object look *smaller* but also get *dimmer* in negatively curved space? According to classical physics, we expect a smaller size to be proportional to the intensity of light. The issue, however, is that the geodesics that the light travel *diverges*.

For example, suppose we are interested in the 1D angular size of a light source. This depends on the divergence of only *two* light paths (top and bottom of the object). These light paths spread apart, and so the object looks smaller. On the other hand, consider the 2D brightness of an object. This depends on how the *total* light spreads out over a 2D surface. Since all paths diverge, the surface area of the light sphere grows faster than in flat space, so the object appears dimmer to an observer.

We can take this result to its limiting case where the geodesic distance is much smaller than the radius of curvature. That is, $r \ll R$. By Taylor expanding $\sinh(x) \approx x$ and $\sin(x) \approx x$, the function $S_\kappa(r)$ simply becomes

$$S_\kappa(r) \approx r \quad (22)$$

for all $\kappa = -1, 0, 1$. In fact, our general space metric becomes the flat-space metric,

$$(dl)^2 \approx (dr)^2 + r^2(d\Omega)^2$$

suggesting that any curved space looks flat on small enough scales. Alternatively, we may wish to represent the space metric with other coordinates. Consider a new circumferential radius $x \equiv S_\kappa(r)$. We can, for example, rewrite the metric for the hyperbolic space case. Here, we now have $x = R \sinh(r/R)$, and taking the differential,

$$dx = \cosh\left(\frac{r}{R}\right) dr \implies dr = \frac{dx}{\cosh(r/R)}.$$

We can use this to eliminate the coordinate r by the identity $\cosh^2(x) - \sinh^2(x) = 1$, so that

$$\cosh\left(\frac{r}{R}\right) = \sqrt{1 + \sinh^2\left(\frac{r}{R}\right)} = \sqrt{1 + \frac{x^2}{R^2}}$$

and so our metric becomes

$$(dl)^2 = \frac{(dx)^2}{1 + (x/R)^2} + x^2(d\Omega)^2.$$

That is, we can represent the same geometry with different coordinate systems.

3.2 The Robertson-Walker Metric

The Robertson-Walker metric describes a spacetime that is spatially homogeneous and isotropic, and allows for expansion and contraction. From the Minkowski metric for Special Relativity, we can include the effects of gravity. In particular, the Robertson-Walker metric takes the form

$$(ds)^2 = -(c dt)^2 + a(t)^2[(dr)^2 + S_\kappa(r)^2(d\Omega)^2]. \quad (23)$$

Here, $a(t)$ is the scale factor describing the expansion of space, which we normalize to be $a(t_0) = 1$ for present time; t is the cosmic time measured by an observer moving with the Hubble flow, and (r, θ, ϕ) are co-moving coordinates which remain fixed as the universe expands (this is important for

the $S_\kappa(r)$ term). In three parameters, we can describe geometry with κ (the curvature constant), R_0 (the present-day radius of curvature), and $a(t)$.

We define the proper distance between two points at a fixed time, t , to be $d_p(t)$.

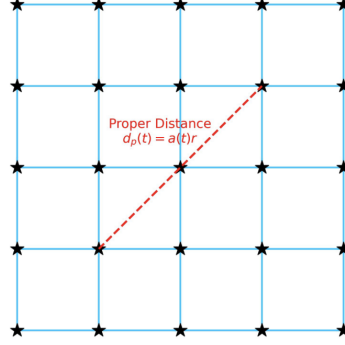


Figure 4: Snapshot of the universe at $a(t) = 0.8$ with a proper distance labelled.

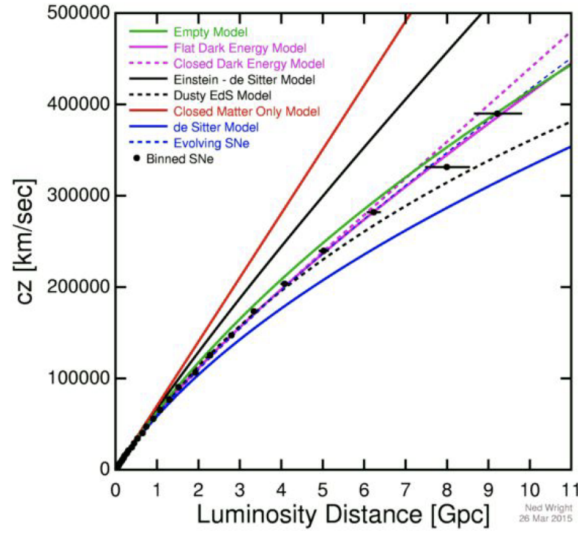


Figure 5: Hubble's Law.

To derive the proper distance, we start with the Robertson-Walker metric at a fixed time $dt = 0$ for a radial path, $d\Omega = 0$. Then,

$$(ds)^2 = a(t)^2(dr)^2 \implies ds = a(t) dr.$$

Taking the integral from the observer at $r = 0$ to some galaxy at a distance r , we have

$$d_p(t) = \int_0^r a(t) dr' = a(t)r.$$

Now, taking the time derivative of the proper distance, we find the proper velocity

$$v_p = \dot{d}_p(t) = \dot{a}r = \dot{a} \left(\frac{d_p}{a} \right) = \left(\frac{\dot{a}}{a} \right) d_p.$$

Let $H(t) = \dot{a}/a$ be the Hubble Parameter, then

$$v_p(t_o) = H_0 d_p(t_o)$$

where we say that $t = t_0$, $a(t_0) = 1$ for Hubble's Law.

Notice the implication of Hubble's Law. On the surface, it says that galaxies with a proper distance greater than the Hubble distance, $d_H = c/H_0$, have a recession velocity greater than the speed of light. This is **not a problem!** In fact, since the speed limit of c applies only to motion *within* a static spacetime, the expansion of space is free to move at speeds $v_p \gg c$.

3.3 Redshift, the Scale Factor, and Friedmann

We define the redshift of light, z , to be the stretching of light's wavelength from distance objects. Redshift is directly related to the scale factor of the universe at the time the light was emitted, by

$$z \equiv \frac{\lambda_{\text{obs}} - \lambda_{\text{emit}}}{\lambda_{\text{emit}}}. \quad (24)$$

From the Robertson-Walker Metric again, since light travels on null geodesics, we let $ds = 0$. Again, let $d\Omega = 0$ since light travels radially so that

$$0 = -c^2(dt)^2 + a(t)^2(dr)^2 \implies dr = c \frac{dt}{a(t)}.$$

The co-moving distance, r , is the same between two successive wave crests, so we can integrate over the time between wave crests for emitted and observed time,

$$r_{\text{crest 1}} = r_{\text{crest 2}} \implies \int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{t_e + \lambda_e/c}^{t_o + \lambda_o/c} \frac{dt}{a(t)}.$$

Since $a(t)$ remains constant over a single wave period, we can simplify this to

$$\frac{\lambda_e}{a(t_e)} = \frac{\lambda_o}{a(t_o)}.$$

So, we can use this result to find that

$$1 + z = \frac{a(t_o)}{a(t_e)} = \frac{1}{a(t_e)}. \quad (25)$$

This relationship between redshift and the scale factor allows us to use our telescopes as "time machines". In fact, for example, a galaxy with a redshift of $z = 3$ would be seen as it was when the universe was 1/4th its current size.

We begin the discussion of the Friedmann Equation by prefacing that we utilize a Newtonian approach. More rigorously, we would (as Friedmann did) begin with the Einstein Field Equation, which would involve solving ten coupled, non-linear second-order PDEs, which is horrible. Thus, from a Newtonian energy conversation standpoint, we will make three assumptions and correct them later. First, we assume that the universe has a centre (homogeneity violation); we next assume a static, Euclidean space (GR violation); we finally assume that gravity is independent of pressure.

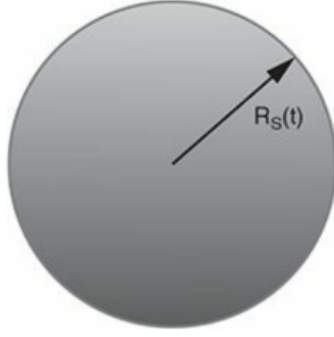


Figure 6: Homogeneous sphere of mass M_s with a central radius $R_s(t)$, free to expand or contract in time.

To begin the derivation by considering a homogeneous sphere of matter and total mass, M_s , constant in time. Assume that the sphere is free to contract or expand isotropically, and so a test mass, m , at a distance from the centre, $R_s(t)$, experiences a gravitational force

$$F = -\frac{GM_s m}{R_s(t)^2}.$$

By Newton's Second Law, we thus have

$$\frac{d^2 R_s}{dt^2} = -\frac{GM_s}{R_s(t)^2} \implies \left(\frac{dR_s}{dt}\right) \left(\frac{d^2 R_s}{dt^2}\right) = -\left(\frac{GM_s}{R_s(t)^2}\right) \left(\frac{dR_s}{dt}\right)$$

where we have multiplied through by dR_s/dt . Notice that both sides are a result of the chain rule, and so we have

$$\frac{1}{2} \frac{d}{dt} \left(\frac{dR_s}{dt}\right)^2 = \frac{d}{dt} \left(\frac{GM_s}{R_s}\right).$$

Integrating with respect to time, we find

$$\frac{1}{2} \left(\frac{dR_s}{dt}\right)^2 = \frac{GM_s}{R_s(t)} + U$$

for some constant of integration (which is an energy term), U . Notice that this energy conservation statement says that the kinetic energy and gravitational potential energy per unit mass, m , is constant for a bit of matter at the surface of the sphere, regardless of expansion or contraction. We assumed that M_s is constant, and so we may write that

$$M_s = \frac{4}{3}\pi\rho(t)R_s(t)^3.$$

About the centre, the expansion is isotropic, so we also have $R_s(t) = a(t)r_s$ for the scale factor $a(t)$, and thus we have

$$\frac{1}{2}r_s^2\dot{a}^2 = \frac{4\pi G r_s^2}{3}\rho(t)a(t)^2 + U.$$

Dividing through by $r_s^2 a^2/2$, we find

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho(t) + \frac{2U}{r_s^2} \frac{1}{a(t)^2},$$

which is the **Friedmann Equation** in the Newtonian form. Let us make a few remarks. Notice that the case $\dot{a} > 0$ is effectively identical to the case $\dot{a} < 0$. The type of expansion thus depends instead on the constant, U . If $U > 0$, then the RHS is positive and $\dot{a}^2 > 0$ always. We conclude that the sphere *expands unboundedly*. If $U < 0$, then for some U , we have a maximum scale factor

$$a_{\max} = -\frac{GM_s}{Ur_s}$$

at which point the RHS is zero and so the expansion stops. Then, since $\ddot{a} < 0$, we conclude that for all remaining time, the sphere contracts. The last boundary case is when $U = 0$. Here, we simply have $\dot{a} \rightarrow 0$ as $t \rightarrow 0$ and $\rho \rightarrow 0$, which isn't particularly interesting.

We can now apply two corrections to arrive at the proper Friedmann Equation. In GR, we know that all forms of energy are related by

$$E^2 = m^2 c^4 + p^2 c^2$$

so for non-relativistic matter, we have the approximations $E \approx mc^2$ and $\epsilon \approx \rho c^2$. Of course, for relativistic particles, we have the relation $E = pc = hf$, so we must account for this by replacing mass density with total energy density. That is, we let

$$\rho(t) \rightarrow \frac{\epsilon}{c^2}.$$

For the second correction, we must connect the dynamics of the universe's geometry to the expansion of space itself. Here, we let the Newtonian energy constant, U , be replaced by the geometric curvature constant, κ . Thus, we have a direct link with our different cases of U : bound ($U < 0 \iff \kappa = 1$), unbound ($U > 0 \iff \kappa = -1$), and critical ($U = 0 \iff \kappa = 0$). Thus, we shall replace

$$\frac{2U}{r_s^2} \rightarrow -\frac{\kappa c^2}{R_0^2}.$$

With these two substitutions, we arrive at the **correct Friedmann Equation**,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2} \epsilon(t) - \frac{\kappa c^2}{R_0^2 a(t)^2}. \quad (26)$$

Notice that the proper representation reveals a remarkable result for the flat universe. Here,

$$H(t)^2 = \frac{8\pi G}{3c^2} \epsilon(t).$$

That is, for any given expansion rate $H(t)$, there is some unique **critical density**, $\epsilon_c(t)$, required to make the universe flat. We denote this by

$$\epsilon_c(t) \equiv \frac{3c^2 H(t)^2}{8\pi G}. \quad (27)$$

We can use the critical density as a reference point for a dimensional quantity called the **density parameter**,

$$\Omega(t) \equiv \frac{\epsilon(t)}{\epsilon_c(t)}. \quad (28)$$

In this way, we can rewrite the Friedmann equation in the form

$$1 - \Omega(t) = -\frac{\kappa c^2}{R_0^2 a(t)^2 H(t)^2}.$$

So, we can directly link the connections between our three universal geometries with Ω : closed ($\Omega > 1 \iff \kappa = 1$), open ($\Omega < 1 \iff \kappa = -1$), and flat ($\Omega = 1 \iff \kappa = 0$).

Suppose we wish to calculate the critical density of the universe in the present time using the Hubble parameter $H_0 = 68 \text{ kms}^{-1}\text{Mpc}^{-1}$. In SI units, we can find that the Hubble parameter is approximately $H_0 = 2.20 \cdot 10^{-18} \text{ s}^{-1}$. We need only to plug into the formula for $\epsilon_c(t = t_0)$, which is

$$\epsilon_c(t_0) = \frac{3c^2 H_0^2}{8\pi G} = 7.8 \cdot 10^{-10} \text{ Jm}^{-3}.$$

In physical units, we reconvert this to find that $\epsilon_c(t_0) = 4870 \text{ MeVm}^{-3}$ and $\rho_c(t_0) \approx 5$ protons per m^3 .

3.4 The Fluid and Acceleration Equations

Unfortunately, since we have two time-dependent quantities ($\epsilon(t)$, $a(t)$), we need two other equations to understand how the scale factor $a(t)$ evolves with time. One of the equations which will help is, eventually, is the *acceleration equation*. In the process of deriving this, an equation which helps us (and is also in the same vein as energy conservation) is the first law of thermodynamics, which, if you recall, is of the form

$$dQ = dE + P dV.$$

Since the universe is homogeneous, its expansion is adiabatic, and so we have $dQ = 0$. Physically, this means that the change in internal energy is balanced by the fluid's pressure as the volume expands. Dividing the first law by the differential time, we thus have

$$\dot{E} + P\dot{V} = 0.$$

As before, let us consider a sphere co-moving radius, r_s , expanding with the universe such that we have the proper radius $R_s(t) = a(t)r_s$. We have the volume

$$V(t) = \frac{4\pi}{3} r_s^3 a(t)^3$$

and so the rate of change (by the chain rule) nets

$$\dot{V} = \frac{4\pi}{3} r_s^3 (3a^2 \dot{a}) = 3V \left(\frac{\dot{a}}{a} \right).$$

We have that the internal energy of the sphere is defined by $E(t) = V(t)\epsilon(t)$, where $\epsilon(t)$ is the pressure in time, so that the rate of change of the internal energy is

$$\dot{E} = V\dot{\epsilon} + \dot{V}\epsilon = V \left(\dot{\epsilon} + 3\epsilon \frac{\dot{a}}{a} \right).$$

We combine each differential equation to find that the first law in an expanding or contracting universe is of the form

$$V \left(\dot{\epsilon} + 3\epsilon \frac{\dot{a}}{a} + 3P \frac{\dot{a}}{a} \right) = 0$$

or, equivalently,

$$\dot{\epsilon} + 3 \frac{\dot{a}}{a} (\epsilon + P) = 0. \tag{29}$$

We call the above equation the **fluid equation**, which will allow us to see how the scale factor evolves with the help of the Friedmann equation. In fact, we will derive an acceleration equation by taking the time derivative of the Friedmann equation,

$$2\dot{a}\ddot{a} = \frac{8\pi G}{3c^2}(\dot{\epsilon}a^2 + 2\epsilon a\dot{a}).$$

Dividing by $2\dot{a}\ddot{a}$, we find that

$$\frac{\ddot{a}}{\dot{a}} = \frac{4\pi G}{3c^2}\left(\dot{\epsilon}\frac{a}{\dot{a}} + 2\epsilon\right).$$

Now, using the fluid equation, we substitute $\dot{\epsilon}a/\dot{a} = -3(\epsilon + P)$ to find that the **acceleration equation** is of the form

$$\frac{\ddot{a}}{\dot{a}} = -\frac{4\pi G}{3c^2}(\epsilon + 3P). \quad (30)$$

Notice that the sign of $(\epsilon + 3P)$ tells us whether the universe accelerates or decelerates. We assume that $\epsilon > 0$ always, since it must act to slow the expansion of the universe. Thus, we have two important cases. In the first case, we may have normal matter or radiation, where $P \geq 0$. Here, we find that

$$P \geq 0 \implies \ddot{a} < 0$$

so that gravity is attractive (GR shows that pressure induces gravity) and thus the expansion *slows down*. In the second case, we may have that $P < -\epsilon/3$, where

$$P < -\frac{\epsilon}{3} \implies \ddot{a} > 0$$

so that some unobservable form of energy creates repulsive gravity, thereby causing the expansion of the universe to *accelerate*.

3.5 Equation of State

We derive the last needed equation which relates pressure and energy density up to a dimensionless parameter, w . We begin by considering a low-density gas of non-relativistic massive particles. That is, we assume that $v \ll c$. By the ideal gas law, we have the relation

$$P = \frac{\rho}{\mu}kT$$

where μ is the mean mass of the gas particles, and ρ is the number density of the gas. For a non-relativistic gas, we have that the energy density obeys the relation $\epsilon \approx \rho c^2$, which says that ϵ is almost entirely composed of the mass of gas particles. Thus, we have that the ideal gas law is

$$P \approx \frac{kT}{\mu c^2}\epsilon.$$

We cite energy equipartition to show that the mean square thermal velocity, $\langle v^2 \rangle$, is associated with the relation

$$3kT = \mu \langle v^2 \rangle$$

and so we can write the **Equation of State** to be of the form

$$P = w\epsilon, \quad w \approx \frac{\langle v^2 \rangle}{3c^2} \quad (31)$$

and we note that $w \ll 1$. Notice that for different values of w , we have different values of pressure. In fact, unique values of w correspond to the pressure exerted by different matter. For example, we may have

$$P_{\text{dust}} = 0, \quad w = 0$$

for **pressure-less dust**. This may consist of non-relativistic particles like protons, neutrons, and dark matter which constitute cosmological objects like galaxies. They have mass and energy, but the pressure exerted is negligible. Similarly, we may have

$$P_{\text{rel}} = \frac{1}{3}\epsilon_{\text{rel}}, \quad w = \frac{1}{3}$$

for **radiation** matter, like photons or neutrinos. The pressure exerted from these objects come from momentum, which is tied to energy. Finally, we also have pressure exerted by **dark energy**, such that

$$P_{\Lambda} < -\frac{1}{3}\epsilon_{\Lambda}, \quad w < -\frac{1}{3} \quad (32)$$

which arises from a component of the universe which we cannot see. However, we can see from the acceleration equation that this component of matter leads to expansion, since

$$\epsilon + 3P < 0 \implies \frac{\ddot{a}}{a} > 0.$$

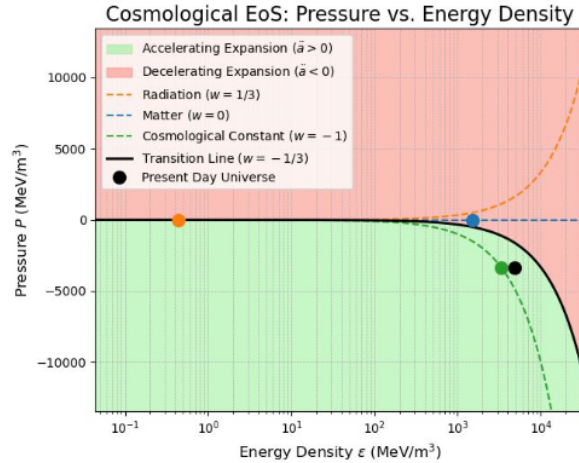


Figure 7: Graphical representation of the components of the universe over a pressure vs. energy density function.

3.6 Energy Evolution and the Cosmological Constant

Now that we have different components of the energy density of the universe,

$$\epsilon = \sum_i \epsilon_i, \quad P = \sum_i P_i = \sum_i w_i \epsilon_i,$$

we can, by assuming that there is no interaction between components, take note that they all individually obey the fluid equation. Using this, we can derive how energy density dilutes as space expands. We begin by starting with the fluid equation for a single component, i ,

$$\dot{\epsilon}_i + 3\frac{\dot{a}}{a}(\epsilon_i + P_i) = 0$$

and substituting the relation $P_i = w_i \epsilon_i$, we rearrange this into a separate differential equation,

$$\dot{\epsilon}_i + 3\frac{\dot{a}}{a}\epsilon_i(1 + w_i) = 0 \implies \frac{d\epsilon_i}{\epsilon_i} = -3(1 + w_i)\frac{da}{a}.$$

Integrating both sides over the bounds (a, a') , we find

$$\ln\left(\frac{\epsilon_i(a)}{\epsilon_i(a')}\right) = -3(1 + w_i)\ln\left(\frac{a}{a'}\right).$$

We let $a' = 1$ to represent the scale factor at the present time, $a' = a(t_0)$, and so taking the exponential, we have

$$\epsilon_i(a) = \epsilon_{i,0}a^{-3(1+w_i)} \quad (33)$$

for $\epsilon_i(a') = \epsilon_{i,0}$. Notice that the exponent, $-3(1 + w_i)$, tells us how each component's energy density dilutes as the scale factor, a , increases. For non-relativistic matter ($w = 0$), we find that

$$\epsilon_m(a) = \frac{\epsilon_{m,0}}{a^3}$$

assuming no particles are formed or destroyed in time. Notice, that since $n \propto a^{-3}$, the energy density of non-relativistic matter dilutes with volume/particle density. For radiation ($w = 1/3$), we find that

$$\epsilon_r(a) = \frac{\epsilon_{r,0}}{a^4}$$

where we also assume that no photons are created or destroyed.³ Since we already know that photon density has the proportionality $n \propto a^{-3}$, we must reference another proportionality. Indeed, the energy per photon as the wavelength expands with the universe has $E = hc/\lambda \propto a^{-1}$, so we can surmise that

$$nE \propto \epsilon \propto a^{-4}.$$

We conclude that the energy density of radiation dilutes rapidly as space expands. If we consider now dark energy ($w = -1$) for a single component of w , we find that

$$\epsilon_\Lambda(a) = \epsilon_{\Lambda,0}$$

which suggests that energy density is constant. That is, as space expands more dark energy appears. This fact leads to a property of self similarity in time for a universe dominated by Λ . Later, we can see that this leads to an exponential growth of $a(t)$.

Switching our discussion to Λ , we can assume a static universe (Einstein forced a static universe in his initial paper) to rewrite the Poisson equation in the form

$$\nabla^2\phi + \Lambda = 4\pi G\rho \quad (34)$$

which is static if $\Lambda = 4\pi G\rho$. If we proceed with this form of the Poisson equation, we find that the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2}\epsilon - \frac{\kappa c^2}{R_0^2 a^2} + \frac{\Lambda}{3} \quad (35)$$

and the acceleration equation becomes

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\epsilon + 3P) + \frac{\Lambda}{3}.$$

³It is worth referencing Ryden for more discussion on this topic.

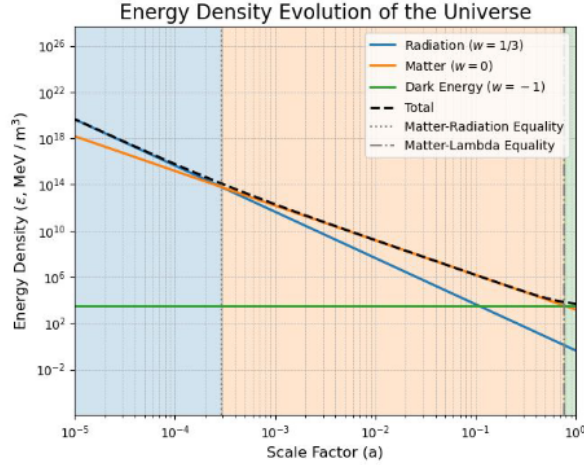


Figure 8: Energy density evolution of the universe as a function of the scale factor.

Notice that this is equivalent to adding a new energy density with the component $w = -1$, where

$$\epsilon_{\Lambda} = \frac{c^2}{8\pi G}\Lambda, \quad P_{\Lambda} = -\epsilon_{\Lambda}.$$

In this way, Einstein got a static universe with matter only and if $\Lambda = 4\pi G\rho/3$. If Λ is larger than this quantity, we end up with the acceleration of the universe, as can be seen by the acceleration equation.

Λ is weird as we do not have a theory giving its explicit value. There are, however, two possible origins of Λ . From QFT, the Heisenberg uncertainty principle allows for the spontaneous creation and annihilation of virtual particle-antiparticle pairs, which would give rise to Λ . On the other hand, we may have some "vacuum energy", ϵ_{vac} , which possesses a constant density regardless of expansion. This would also result in a value of Λ . However, our observations are off by 120 orders of magnitude from the observed volume, making it a major unsolved problem today.

3.6.1 Dark Energy Dominated Universe

Recall that as $a(t)$ evolves, different components of the universe dominate the energy budget of the universe. For radiation density today (from the CMB), where $T_0 = 2.7255$ K, we have that $\epsilon_{\text{CMB},0} = \alpha T_0^4 = 0.2606 \text{ MeV m}^{-3}$ and $\Omega_{\text{CMB},0} \equiv \epsilon_{\text{CMB},0}/\epsilon_{c,0} \approx 5.35 \cdot 10^{-5}$. We also know from observations that $\Omega_{m,0} \approx 0.3$ and $\Omega_{\Lambda,0} \approx 0.7$. These observations tell us we are in a dark energy dominated universe. Is this the case for different time frames of the universe?

Recall that we have the relations $\epsilon_m(a)$, $\epsilon_r(a)$, and ϵ_{Λ} . With these relations, we have different **epochs of equality**. The epoch of matter-radiation equality, we have is

$$\frac{\epsilon_m(a)}{\epsilon_r(a)} = \frac{\Omega_{m,0}}{\Omega_{r,0}} a_{rm} = 1 \implies a_{rm} = 2.9 \cdot 10^{-4}, \quad z_{rm} = \frac{1}{a_{rm}} - 1 \approx 3440. \quad (36)$$

The epoch of matter-lambda equality, we have is

$$\frac{\epsilon_m(a)}{\epsilon_{\Lambda}(a)} = \frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} a_{m\Lambda}^3 = 1 \implies a_{m\Lambda} \approx 0.766, \quad z_{m\Lambda} \approx 0.33. \quad (37)$$

Thus, given a and z are monotonic functions of t , we can use them as a proxy for time. We can use these relations to understand the standard cosmological model today. The full Friedmann equation, describing a universe with matter, radiation, and the cosmological constant and curvature is in the form

$$\frac{H(t)}{H_0^2} = \frac{\Omega_{r,0}}{a(t)^4} + \frac{\Omega_{m,0}}{a(t)^3} + \Omega_{\Lambda,0} + \frac{1 - \Omega_0}{a(t)^2} \quad (38)$$

where $\Omega_{m,0}$, $\Omega_{r,0}$, $\Omega_{\Lambda,0}$ are the density parameters for the different components of the universe. The total density parameter today is thus

$$\Omega_0 = \Omega_{r,0} + \Omega_{m,0} + \Omega_{\Lambda,0} \approx 1$$

which is one according to observations today. The individual components today are thusly

$$\Omega_{m,0} \approx 0.31, \quad \Omega_{r,0} \approx 9 \cdot 10^{-5}, \quad \Omega_{\Lambda,0} \approx 0.69.$$

From Einstein's theory, we can connect curvature and the fate of the universe. Recalling the Friedmann equation in $\Omega(t)$, we can connect each geometry to a different fate of the universe. For $\kappa = 1$, we have a bound system and so gravity is the dominant force, resulting in the **big crunch**. For $\kappa = 0$, we have the gravity is perfectly balanced, but there is a critical density, so we have an **infinite expansion**. Lastly, for $\kappa = -1$, we have an unbound system and so gravity loses as a force, resulting in the **big chill**.

3.7 Model Universes

We will condense the Friedmann equation into the form

$$\dot{a}^2 = \frac{8\pi G}{3c^2} \sum_i \epsilon_{i,0} a^{-1-3w_i} - \frac{\kappa c^2}{R_0^2}.$$

3.7.1 Model 1: The Empty Universe.

The simplest possible model we can consider is the universe with no matter, radiation, or dark energy. We have $\epsilon_i = 0$, and so the Friedmann equation simplifies to

$$\dot{a}^2 = -\frac{\kappa c^2}{R_0^2}.$$

Notice we cannot have $\kappa = 1$, since $\dot{a}^2 < 0$ which is not physically possible. We may have $\kappa = 0$, which gives an empty, flat, and static universe since $\dot{a} = 0$ also. In the special case $\kappa = -1$, we have a constantly expanding or contracting universe, with

$$\dot{a} = \pm \frac{c}{R_0}.$$

Further, we have c/R_0 which is constant, so the empty universe in the hyperbolic space expands and contracts at a constant rate. We can explicitly derive the scale factor for this empty universe. Starting from the Friedmann equation, we have

$$\frac{da}{dt} = \frac{c}{R_0} \implies a(t) = \frac{c}{R_0} t = \frac{t}{t_0}$$

with the boundary conditions $a(0) = 0$ and $a(t_0) = 1$. In particular, the Hubble time is

$$H(t) = \frac{\dot{a}}{a} = \frac{1/t_0}{t/t_0} = \frac{1}{t}$$

which is the age of the universe as well. For the proper distance, we consider the time of emission, t_e , as before, which has the redshift relation

$$t_e = \frac{t_0}{1+z}.$$

Integrating along a photon's null geodesic, we have

$$d_p(t_0) = \int_{t_e}^{t_0} \frac{c dt}{a(t)} = \int_{t_e}^{t_0} \frac{ct_0}{t} dt = \frac{c}{H} \ln(1+z)$$

where we have substituted $a(t) = t/t_0$. Similarly, we can write the proper distance as an integral over z or a , where we have the relations

$$d_p(t_0) = \int_{t_e}^{t_0} \frac{c dt}{a(t)} = \int_{a_e}^1 \frac{a da}{a^2 H(a)} = \int_0^z \frac{c dz}{H(z)}.$$

For a multi-component universe, solving for $a(t)$ can be mathematically intractable. However, we may be able to transform this integral to be over other variables such as the scale factor itself, a , or the redshift, z , which is valid for any Friedmann model.

Applying these observations to cosmic history, the age of the universe at some given redshift z is represented by

$$t(z) = \int_0^\infty \frac{dz'}{c(1+z')H(z')}$$

and a numerical solution needs only

$$H_0 t = \int_0^a \frac{da'}{\sqrt{\Omega_{r,0}/(a')^2 + \Omega_{m,0}/a' + \Omega_{\Lambda,0}(a')^2 + (1 - \Omega_0)}}.$$

Based on the **Benchmark Model**, we can look at the time and redshift of key cosmic events.

- **Matter-Radiation Equality:**

- Scale Factor: $a_{rm} \approx 2.9 \cdot 10^{-4}$
- Redshift: $z_{rm} \approx 3440$
- Time: $t_{rm} \approx 50,000$ years

- **Cosmic Microwave Background (CMB):**

- $z_{ls} \approx 1100$
- $t_{ls} \approx 370,000$ years after the Big Bang

- **Matter-Dark Energy Equality:**

- Scale Factor: $a_{m\Lambda} \approx 0.766$
- Redshift: $z_{m\Lambda} \approx 0.33$
- Time: $t_{m\Lambda} \approx 9.8$ billion years.

3.7.2 Model 2: Flat, Single-Component Universes

Let us assume now we have a flat universe with parameters ($\kappa = 0, \Omega_0 = 1$) with a single component. Here, the Friedmann equations takes the form

$$\dot{a}^2 = \frac{8\pi G\epsilon_0}{3c^2} a^{-(1+3w)}.$$

We derive the scale factor for this model universe by assuming a power-law solution of the form $a = (t/t_0)^q$ and substituting it into the Friedmann equation. In this way, we have

$$\frac{q^2}{t_0^2} \left(\frac{t}{t_0}\right)^{2(q-1)} = \frac{8\pi G\epsilon_0}{3c^2} \left(\frac{t}{t_0}\right)^{-q(1+3w)}.$$

For this to be valid, we equate the powers of (t/t_0) so that we find

$$q = \frac{2}{3(1+w)}, \quad (w \neq -1).$$

Of course, this gives the scale factor proportionality $a(t) \propto t^{2/[3(1+w)]}$. Indeed, we find

$$a = \left(\frac{t}{t_0}\right)^{\frac{2}{3(1+w)}} \quad \text{where } t_0 = \frac{1}{1+w} \frac{c}{\sqrt{6\pi G\epsilon_0}}.$$

The age of the universe is given in terms of the Hubble time, which would be

$$t_0 = \frac{2}{3(1+w)} H_0^{-1}.$$

Notice, if $w > -1/3$, the universe is younger than the Hubble time since $t_0 < H_0^{-1}$. However, if $w < -1/3$, the universe is older than the Hubble time since $t_0 > H_0^{-1}$. If we wish to express the Hubble parameter as a function of redshift, we have the relation

$$H(z) = H_0(1+z)^{\frac{3}{2}(1+w)}.$$

Indeed, we can compute the proper distance to a galaxy within this universe by considering the time emitted, which would be

$$t_e = \frac{t_0}{(1+z)^{3(1+w)/2}} = \frac{2}{3(1+w)H_0} \frac{1}{(1+z)^{3(1+w)/2}}$$

where we have used $1+z = a$. The proper distance is thus

$$d_p(t_0) = \int_{t_e}^{t_0} \frac{c dt}{a(t)} = ct_0 \frac{3(1+w)}{1+3w} \left[1 - \left(\frac{t_e}{t_0}\right)^{(1+3w)/[3(1+w)]} \right]$$

for $w \neq -1/3$. We rewrite this in terms of H_0 and z rather than t_0 and t_e , so that we have

$$d_p(t_0) = \frac{2c}{H_0(1+3w)} \left[1 - (1+z)^{-\frac{1}{2}(1+3w)} \right].$$

Taking the limiting case $z \rightarrow \infty$, we get the horizon distance which would be

$$d_{\text{hor}}(t_0) = \frac{2c}{H_0(1+3w)}.$$

Notice, if $w \leq -1/3$, we have that $d_{\text{hor}}(t_0) \rightarrow \infty$ suggesting that all of the universe can be seen. Unsurprisingly, if $w > 1/3$, the horizon distance is finite.

Matter-Only Universe. We shall consider a matter only universe which is the case $w = 0$. Here, we have a universe with only pressure-less matter, and so parameters

$$t_0 = \frac{2}{3}H_0^{-1}, \quad a(t) = \left(\frac{t}{t_0}\right)^{\frac{2}{3}}, \quad d_{\text{hor}} = 2cH_0^{-1}, \quad d_p(t_0) = 2c\left[1 - \frac{1}{\sqrt{1+z}}\right]H_0^{-1}.$$

In this model, we find that expansion slows down due to gravity and that the universe is younger than the Hubble time.

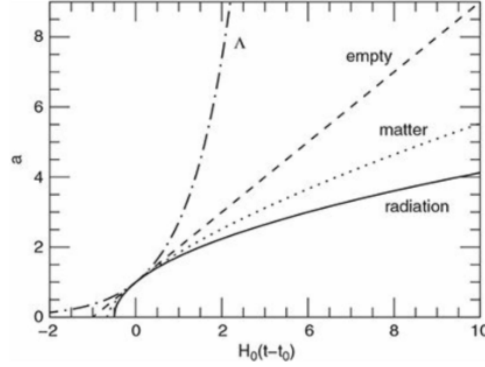


Figure 9: Three different cases of matter universes as the Hubble time increases.

Explicitly, we derive the proper distance such that

$$d_p(t_0) = \int_{t_e}^{t_0} \frac{c dt}{(t/t_0)^{2/3}} = 3ct_0 \left[1 - \left(\frac{t_e}{t_0}\right)^{1/3}\right] = \frac{2c}{H_0} \left[1 - \frac{1}{\sqrt{1+z}}\right]$$

as before. At the time of emission, we pick up a scaling factor of $1/(1+z)$, so that

$$d_p(t_e) = \frac{2c}{H_0(1+z)} \left[1 - \frac{1}{\sqrt{1+z}}\right].$$

Radiation-Only Universe. We consider the radiation-only model which has the parameter $w = 1/3$. Correspondingly, our general solution for q is $q = 1/2$, and so we have $a(t) \propto t^{1/2}$. The age of the universe promptly follows

$$t_0 = \frac{1}{2}H_0^{-1}$$

with a horizon distance

$$d_{\text{hor}}(t_0) = \frac{c}{H_0}.$$

Similarly, the proper distance follows and is

$$d_p(t_0) = \frac{c}{H_0} \frac{z}{1+z}.$$

Lambda-Only Model. We now consider a model which has only Λ with $w = -1$. Here, the power-law solution breaks down, so we return to the Friedmann equation for a flat universe, where

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\epsilon_{\Lambda,0}}{3c^2}.$$

Since $\epsilon_{\Lambda,0}$ is constant with time, we can rewrite this in the form

$$\dot{a} = H_0 a, \quad H_0 = \sqrt{\frac{8\pi G \epsilon_{\Lambda,0}}{3c^2}}.$$

The solution to this differential equation is simple, and is of the form

$$a(t) = e^{H_0(t-t_0)}.$$

In this model, we have an infinitely aged universe, with a proper distance

$$d_p(t_0) = \frac{c}{H_0} z.$$

Correspondingly, the horizon distance is infinite, with $d_{\text{hor}}(t_0) \rightarrow \infty$.

We can compare each single component model with a single table, illustrated in Table 1.

Component	EoS (w)	Scaling (ϵ)	Scale Factor (a)	Age (t_0)	Horizon (d_{hor})
Matter	0	$\propto a^{-3}$	$\propto t^{2/3}$	$(2/3)H_0^{-1}$	$2c/H_0$
Radiation	1/3	$\propto a^{-4}$	$\propto t^{1/2}$	$(1/2)H_0^{-1}$	c/H_0
Lambda	-1	$\propto a^0$	$\propto e^{H_0(t-t_0)}$	∞	∞

Table 1: Summary of cosmological components and their corresponding scaling relations, scale factors, and horizon distances.

3.7.3 Model 3: Multi-Component Universes

To derive the model for the multi-component universe, we begin with the Friedmann equation with $H(t)$,

$$H(t)^2 = \frac{8\pi G}{3c^2} \epsilon(t) - \frac{\kappa c^2}{R_0^2 a(t)^2}.$$

If we recall that we have, for the critical density and the density parameters,

$$\epsilon_{c,0} = \frac{3c^2 H_0^2}{8\pi G} \text{ (critical density)}, \quad \Omega = \frac{\epsilon(t)}{\epsilon_c(t)} \text{ (density parameter)},$$

we can rewrite the present day Friedmann equation to have (after diving by H_0^2),

$$\frac{H(t)^2}{H_0^2} = \frac{\epsilon(t)}{\epsilon_{c,0}} + \frac{1 - \Omega_0}{a(t)^2}.$$

Using this result, we note that $\epsilon(t)$ is composed of matter, radiation, and lambda, so that we have

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3} + \Omega_{\Lambda,0} + \frac{1 - \Omega_0}{a^2} \quad (39)$$

where we have

$$\Omega_{r,0} = \frac{\epsilon_{r,0}}{\epsilon_{c,0}}, \quad \Omega_{m,0} = \frac{\epsilon_{m,0}}{\epsilon_{c,0}}, \quad \Omega_{\Lambda,0} = \frac{\epsilon_{\Lambda,0}}{\epsilon_{c,0}}, \quad \Omega_0 = \Omega_{r,0} + \Omega_{m,0} + \Omega_{\Lambda,0}.$$

This model is unique as there is no simple analytic solution to many of our observable quantities. As an example, consider the age given by the integral in the scaling factor,

$$H_0 t = \int_0^a \frac{da}{\sqrt{\Omega_{r,0}/a^4 + \Omega_{m,0}/a^3 + \Omega_{\Lambda,0} + (1 - \Omega_0)/a^2}}.$$

However, we may make approximations where one component is negligible. Suppose, for example, we have the historical model where there is only matter and curvature. That is, we have $\Omega_{m,0} = \Omega_0$ to give

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_0}{a^3} + \frac{1 - \Omega_0}{a^2}. \quad (40)$$

Using the Newtonian analogy, we can interpret this equation in terms of the energy density. If, for example, the total energy density was greater than the critical density, we would have $\Omega_0 > 1$ so that the universe is positively curved. If the total energy density was less than the critical density, we would have $\Omega_0 < 1$ and the universe becomes negatively curved.

Closed Universes. We will consider the closed universe in the case $\Omega_0 > 1$, the **big crunch**. Here, we have a maximum expansion point where $H_0 = 0$. The scale factor at this point is correspondingly

$$a = a_{\max} = \frac{\Omega_0}{\Omega_0 - 1}.$$

This solution is time-reversible, but note that entropy does *not* reverse. We can further derive a parametric solution for $a(t)$, which becomes of the form

$$\begin{aligned} t(\theta) &= \frac{1}{2H_0} \frac{\Omega_0}{(\Omega_0 - 1)^{3/2}} [\theta - \sin(\theta)] \\ a(\theta) &= \frac{\Omega_0}{2(\Omega_0 - 1)} [1 - \cos(\theta)]. \end{aligned}$$

Notice that after reaching a maximum expansion, which is the case $\theta = 2\pi$, the universe begins to contract and enter a "big crunch".

Open Universes. Now consider an open universe in the case $\Omega_0 < 1$. Here, both terms in the Friedmann equations are position, so we have an infinite expansion. This specific model shows that we begin with matter-dominated space, where $a \propto t^{2/3}$, but at some point becomes curvature-dominated with $a \propto t$. Correspondingly, our parametric solutions for $a(t)$ are

$$\begin{aligned} t(\eta) &= \frac{1}{2H_0} \frac{\Omega_0}{(1 - \Omega_0)^{3/2}} [\sinh(\eta) - \eta] \\ a(\eta) &= \frac{\Omega_0}{2(1 - \Omega_0)} [\cosh(\eta) - 1]. \end{aligned}$$

Flat Universes. In the special case where $\Omega_0 = 1$, we have zero curvature for $\kappa = 0$. The solution shows that the expansion rate slows down over time, asymptotically approaching zero.

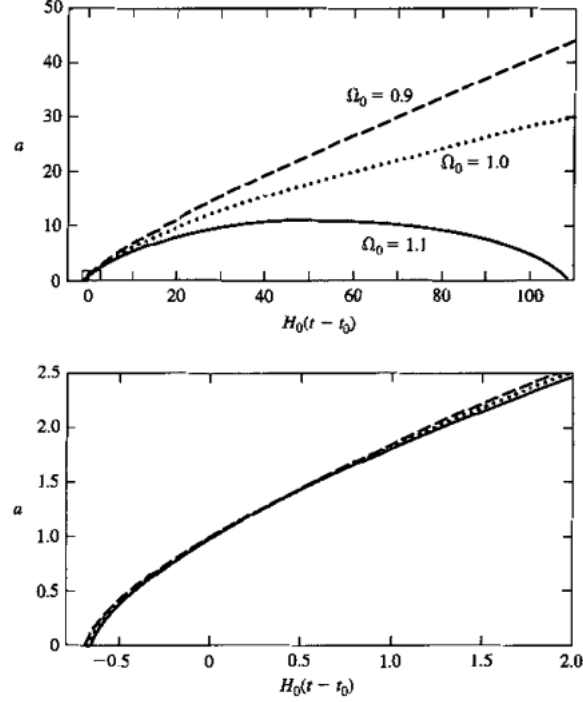


Figure 10: The scale factor as a function of the Hubble time for matter-only universes. $\Omega_0 = 1$ corresponds to the dotted line, $\Omega_0 = 0.9$ corresponds to the dashed line, and $\Omega_0 = 1.1$ corresponds to the solid line.

3.7.4 Model 4: The Matter-Lambda Modern Model

Our next logical step is to reintroduce the cosmological constant. Here, we assume a flat geometry, so that we have observations that align with our theory. The Friedmann equation in this case is of the form

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{m,0}}{a^3} + (1 - \Omega_{m,0}).$$

Of course, we have two major scenarios corresponding to the big crunch ($\Omega_{m,0} > 1$, $\Omega_{\Lambda,0} < 0$) which aligns with a universe with an attractive cosmological constant, and the big chill ($\Omega_{m,0} < 1$, $\Omega_{\Lambda,0} > 0$) which has observations which fit the universe.

The Big Crunch. Here, we have a negative cosmological constant. We can derive the expansion to show that there is a maximum scale factor

$$a_{\max} = \left(\frac{\Omega_{m,0}}{\Omega_{m,0} - 1} \right)^{\frac{1}{3}}$$

to which the universe eventually re-collapses at a cosmic time

$$t_{\text{crunch}} = \frac{2\pi}{3H_0} \frac{1}{\sqrt{\Omega_{m,0} - 1}}.$$

We can derive explicitly the parametric solution, which is of the form

$$H_0 t = \frac{2}{3\sqrt{\Omega_m - 1}} \sin^{-1} \left[\left(\frac{a}{a_{\max}} \right)^{\frac{3}{2}} \right].$$

The Big Chill. This model fits our universe, where the cosmological constant has a dominating presence at some later time. Expansion in this case never stops, with a matter-lambda equality at some scale factor

$$a_{m\Lambda} = \left(\frac{\Omega_m}{1 - \Omega_m} \right)^{\frac{1}{3}}.$$

If we integrate the Friedmann equation, we can yield explicitly the form

$$H_0 t = \frac{2}{3\sqrt{1 - \Omega_m}} \ln \left[\left(\frac{a}{a_{m\Lambda}} \right)^{\frac{3}{2}} + \sqrt{1 + \left(\frac{a}{a_{m\Lambda}} \right)^3} \right].$$

The age of the universe occurs at the point $a = 1$, so that

$$t_0 = \frac{2H_0^{-1}}{3\sqrt{1 - \Omega_m}} \ln \left(\frac{\sqrt{1 - \Omega_m} + 1}{\sqrt{\Omega_m}} \right) = 13.74 \text{ Gyr (for } \Omega_m = 0.31, \Omega_\Lambda = 0.69).$$

By choosing different values for each components, we can get a variety of different cosmic histories. So, we arrive at the full form of the Friedmann equation,

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{m,0}}{a^3} + (1 - \Omega_{m,0} - \Omega_{\Lambda,0})a^{-2} + \Omega_{\Lambda,0} \quad (41)$$

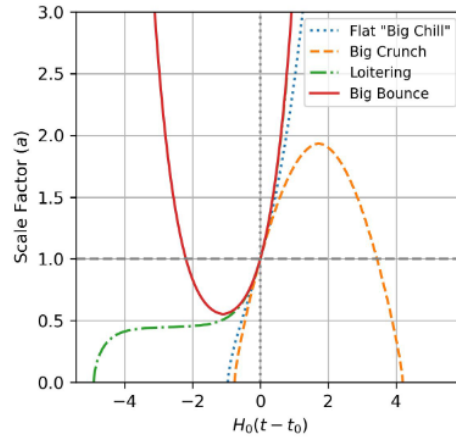


Figure 11: Comparison of the scale factors for each type of universal scenario.

Fig. 12 also shows the landscape of possible universes. The dashed line corresponds to the case where $\Omega_m + \Omega_\Lambda = 1$, and the universe to the left of that line are open, which those to the right are closed.

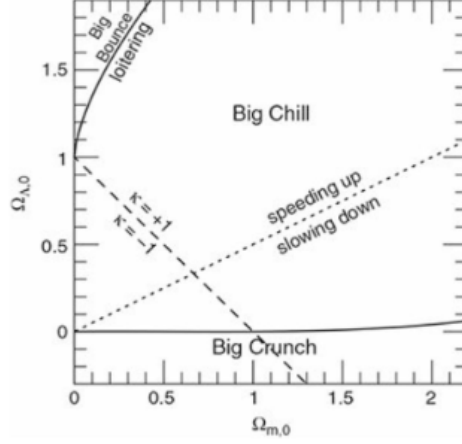


Figure 12: Landscape of possible universes.

We can use this figure to understand how each universe acts over time. The "big crunch" is a re-collapse of the universe, while the "big chill" is an eternal expansion. The "loitering" universe is a unique cases where the universe has a period of stasis, while the "big bounce" is an alternative to the "big band". Interestingly, the "big bounce" has a maximum redshift, so more distance sources would be blue-shifted.

Suppose now we wish to look at the early universe when matter and radiation were in competition for domination. The Friedmann equation for this case is of the form

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3}.$$

Here, we can see that the radiation term dilutes faster with the scale factor (a^{-4}) than the matter term (a^{-3}) in Fig. 13.

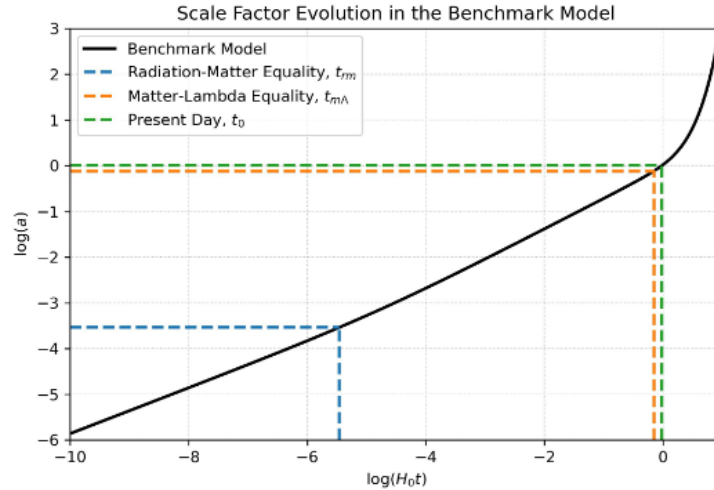


Figure 13: The scale factor evolution in the Benchmark Model.

On another hand, the transition from a radiation-dominated universe to a matter-domination universe is a very important cosmic milestone. We know that this matter-radiation takes place

when

$$a_{\text{rm}} = \frac{\Omega_{r,0}}{\Omega_{m,0}} = 2.9 \cdot 10^{-4}, \quad z_{\text{rm}} \approx 3440.$$

In fact, we have the solution for the cosmic time

$$H_0 t = \frac{4a_{\text{rm}}^2}{3\sqrt{\Omega_{r,0}}} \left[1 - \left(1 - \frac{a}{2a_{\text{rm}}} \right) \left(1 + \frac{a}{a_{\text{rm}}} \right)^{\frac{1}{2}} \right]$$

where the expansion before this equality corresponds to $a \ll a_{\text{rm}}$ with

$$a \approx \left(2\sqrt{\Omega_{r,0}} H_0 t \right)^{\frac{1}{2}}.$$

After, we have the matter-dominated era with $a \gg a_{\text{rm}}$ with

$$a \approx \left(\frac{3}{2} \sqrt{\Omega_{m,0}} H_0 t \right)^{\frac{2}{3}}.$$

The time of radiation-matter equality, t_{rm} , can be found if we set $a = a_{\text{rm}}$, so that

$$t_{\text{rm}} = \frac{4}{3} \left(1 - \frac{1}{\sqrt{2}} \right) \frac{a_{\text{rm}}^2}{\sqrt{\Omega_{r,0}}} H_0^{-1} \approx 0.391 \frac{\Omega_{r,0}^{3/2}}{\Omega_{m,0}^2} H_0^{-1}$$

for the parameters

$$\Omega_{m,0} = 0.31, \quad \Omega_{r,0} = 9.0 \cdot 10^{-5}, \quad t_{\text{rm}} = 50000 \text{ years}, \quad H_0^{-1} = 14.4 \text{ Gyr}.$$

However, note that this era was so brief that its effect on the total age of the universe can be justifiably ignored.

3.7.5 The Benchmark Model

We can currently fit all observations into a single cosmological model. For a flat space, $\kappa = 0$, we have the total density parameter

$$\Omega_0 = \sum_i \Omega_i = 1$$

with a Hubble constant $H_0 = 68 \text{ kms}^{-1} \text{Mpc}^{-1}$. We have radiation components due to photons and neutrinos only, where the photons are black body and the CMB photons have a temperature $T = 2.7255 \text{ K}$. We have that Baryonic matter constitutes $\sim 15\%$ of total matter, with most of it dark, and the current energy density is dominated by dark energy/vacuum energy, equivalent in form to a cosmological constant ($w = -1$).

We summarize briefly the properties of the Benchmark Model.

- Total Radiation: $\Omega_{r,0} = 9.0 \cdot 10^{-5}$
- Baryons: $\Omega_{b,0} = 0.048$
- Dark Matter: $\Omega_{\text{dm},0} = 0.262$
- Total Matter: $\Omega_m = 0.31$

- Cosmological Constant: $\Omega_\Lambda = 0.69$
- Age of the Universe is 13.74 Gyr
- Horizon distance is $d_{\text{hor}} = 14$ Gpc
- Galaxies at $z = 10$ have light emitted at
 1. $t_0 - t_e \approx 13.27$ Gyr ago
 2. $d_p \approx 13.5$ Gpc
- For Comparison, CMB $d_p \approx 14$ Gpc.

We thus see three distance epochs of domination. The **radiation domination**, where the universe expanded as $a(t) \propto t^{1/2}$ until $t_{rm} \approx 50000$ years; the **matter domination** which occurred after radiation diluted, so that the universe expanded as $a(t) \propto t^{2/3}$ for most of cosmic history; the **lambda domination**, where the universe in present time leads to accelerated expansion.

4 Cosmological Parameters

4.1 Kinematics of the Scale Factor

Rather than determining the exact functional form of $a(t)$, we shall instead Taylor expand about the immediate past and near future. That is, the present moment for very small redshifts. This leads to a "model-free" description of $a(t)$. We begin by expanding about t_0 , so that

$$a(t) = a(t_0) + \left. \frac{da}{dt} \right|_{t=t_0} (t - t_0) + \frac{1}{2} \left. \frac{d^2a}{dt^2} \right|_{t=t_0} (t - t_0)^2 + \dots$$

Applying the definition $a(t_0) = 1$, we also have $\dot{a}(t_0) = H_0 a(t_0) = H_0$ and $\ddot{a}(t_0) = -q_0 H_0^2$. Indeed, we define

$$q_0 \equiv -\frac{\ddot{a}(t_0)}{H_0^2}$$

so that we have, to second order in t ,

$$a(t) \approx 1 + H_0(t - t_0) - \frac{1}{2} q_0 H_0^2 (t - t_0)^2. \quad (42)$$

We should be careful in understanding how q_0 interacts with the acceleration of a . If $q_0 > 0$, we have that $\ddot{a}_0 < 0$ so that the universe decelerates (gravity wins). If $q_0 < 0$, we have that $\ddot{a}_0 > 0$ so that the universe accelerates (and dark energy wins).

In writing this Taylor expansion, we can substitute $a(t)$ into the acceleration equation to find that

$$q_0 = \Omega_{r,0} + \frac{1}{2} \Omega_{m,0} - \Omega_{\Lambda,0} = \frac{1}{2} \sum_i \Omega_{i,0} (1 + 3w_i) \quad (43)$$

which is a sum of attractive and repulsive terms. Table 2 summarizes the different components, Ω_i , that contribute to q_0 .

Component (Ω_i)	Equation of State (w)	Contribution to q_0	Effect on Expansion
Radiation ($\Omega_{r,0}$)	$\frac{1}{3}$	$+\Omega_{r,0}$	Deceleration
Matter ($\Omega_{m,0}$)	0	$+0.5\Omega_{M,0}$	Deceleration
Lambda ($\Omega_{\Lambda,0}$)	1	$-\Omega_{\Lambda,0}$	Acceleration

Table 2: Comparison of the different components of Ω_i which contribute to the overall effect of q_0 and the expansion of the universe.

Suppose now we wish to check if a distant galaxy's distance matches the predictions of H_0 and q_0 . A major issue, however, is that the proper distance is *not measurable* in an expanding universe. To solve this, we shall define two quantities based on *observable effects* of light in a general (Robertson-Walker) spacetime: the **luminosity distance** (d_L) from the flux of a **standard candle** (L), and the **angular-diameter distance** (d_A) from the angle of a **standard ruler** (l).

In the Robertson-Walker metric, the scale factor is responsible for the redshift/time-dilation effects on d_L and d_A . Similarly, the curvature term governed by $S_\kappa(r)$ affects the angular size which we observe.

We define the standard candle as a source with a known intrinsic luminosity. The luminosity distance of the standard candle is defined as

$$d_L \equiv \sqrt{\frac{L}{4\pi f}}. \quad (44)$$

Cosmologically, the flux arising from the luminosity have three effects: *geometric spreading*, where light spread over $A_p(t_0) \propto S_\kappa(r)^2$, *energy dilution*, where each photon's energy drops by a factor $1/(1+z)$, and *time dilution*, where the arrive rate of photons drops also by $1/(1+z)$. As a result, we have an observed flux of the form

$$f = \frac{L}{4\pi S_\kappa(r)^2(1+z)^2}. \quad (45)$$

To find an expression for the luminosity distance in a general FRW universe, we substitute the flux expression into d_L ,

$$d_L = \sqrt{\frac{L}{4\pi} \frac{4\pi S_\kappa(r)^2(1+z)^2}{L}} = S_\kappa(r)(1+z).$$

Assuming we have flat geometry,

$$d_L = r(1+z) = d_p(t_0)(1+z)$$

which shows that the simple inverse square law overestimates the true distance by a factor of $(1+z)$. To fix this issue, we resort to the angular-diameter distance, d_A , defined by

$$d_A \equiv \frac{l}{\delta\theta}$$

which requires a standard ruler (an object of known physical length, l). In the Robertson-Walker metric, the proper length is simply

$$l = ds = a(t_e)S_\kappa(r)\delta\theta$$

assuming a null geodesic (so $dr = 0$ and $dt = 0$). Thus, we have

$$d_A = \frac{a(t_e)S_\kappa(r)\delta\theta}{\delta\theta} = a(t_e)S_\kappa(r).$$

We substitute the scale factor relation to redshift, $a(t_e) = 1/(1+z)$, so that

$$d_A = \frac{S_\kappa(r)}{1+z}.$$

Now, we need only to relate d_A and d_L . We know $d_L \propto (1+z)$ and $d_A \propto 1/(1+z)$, so we can eliminate $S_\kappa(r)$ in both terms to find

$$d_L = d_A(1+z)^2 \quad (46)$$

which holds true if there are two valid core principles across the universe: metric compatibility (GR) and photon conservation. Using these results, we can now approximate the theoretical result of the proper distance by a Taylor expansion on $a(t)$. We have that

$$\frac{1}{a(t)} \approx 1 - H_0(t - t_0) + \left(\frac{1+q_0}{2}\right)H_0^2(t - t_0)^2$$

up to second order in t . Integrating this expression over the look back time (t_e, t_0) , we have

$$\begin{aligned} d_p(t_0) &= \int_{t_e}^{t_0} \frac{c \, dt}{a(t)} \\ &= c \int_{t_e}^{t_0} \left[1 - H_0(t - t_0) + \left(\frac{1 + q_0}{2} \right) H_0^2(t - t_0)^2 \right] dt \\ &= c(t_0 - t_e) - \frac{cH_0}{2}(t_0 - t_e)^2 + \mathcal{O}(t^3). \end{aligned}$$

Approximating only up to second order in z , we use the relation $z = 1/a(t) - 1$ so that we have the redshift-look-back time relation,

$$t_0 - t_e \approx H_0^{-1} \left[z - \left(\frac{1 + q_0}{2} \right) z^2 \right].$$

to find that

$$d_p(t_0) \approx \frac{c}{H_0} \left[z - \left(\frac{1 + q_0}{2} \right) z^2 \right] + \frac{cH_0}{2} \frac{z^2}{H_0^2} = \frac{c}{H_0} z \left[1 - \frac{1 + q_0}{2} z \right]. \quad (47)$$

Notice that this result shows that the proper distance is dependent on q_0 since acceleration changes the travel time of light. The next most reasonable question to ask is if we are able to combine $d_p(t_0)$ and d_L to get the observable relation in z . We need only to start with the flat case of d_L , which we know is $d_L = d_p(t_0)(1 + z)$, and after substitutions,

$$d_L = \left(\frac{c}{H_0} z \left[1 - \frac{1 + q_0}{2} z \right] \right) (1 + z) \approx \frac{c}{H_0} z \left[1 - \frac{1 + q_0}{2} z \right] + \frac{c}{H_0} z^2 \left[1 - \frac{1 + q_0}{2} z \right].$$

Approximating to second order in z , we find

$$d_L \approx \frac{c}{H_0} z \left[1 + \frac{1 - q_0}{2} z \right]. \quad (48)$$

Notice that there is a correction term proportional to z^2 . In fact, we let $q_c(z)$ be the correction term such that

$$q_c(z) = -\frac{cq_0}{2H_0} z.$$

Notice, if $q_0 > 0$, then $d_L \downarrow$ since the quantities independent of q_0 are subtracted by a larger q_0 ; we expect the object to appear *brighter* than expected. If $q_0 < 0$, then $d_L \uparrow$ since the quantities are subtracted by a smaller q_0 ; we expect the object to appear *fainter* than expected.

4.2 Variable Measurement

4.2.1 Measuring H_0

We can proceed with measuring H_0 by assuming a linear regime

$$cz \approx H_0 d_L.$$

We need only apply an algorithmic process to determine what H_0 is:

1. Identify a population of standard candles with known luminosity, L .

2. Measure redshift (z) and flux (f) for each standard candle.
3. Compute the luminosity distance for each standard candle.
4. Plot the redshift (cz) against the luminosity distance (d_L).
5. Measure the slope for $z \ll 1$, which gives H_0 .

As an example, consider the Cepheid Variables (CVs), which have a precise **period-luminosity (P-L) relation** ($P = 1.6 \rightarrow 60$ days). We must calibrate the CV P-L relationship with, for example, independent measures of distance. We consider the relative distance between CVs in nearby galaxies. Suppose, for example, we wish to compare M31 and LMC, then the flux ratio gives the distance ratio in a relation

$$\frac{d_L(\text{M31})}{d_L(\text{LMC})} = \sqrt{\frac{f_{\text{LMC}}}{f_{\text{M31}}}}.$$

Using parallax, we perform *absolute calibration* to measure the distance to local CVs, so that we can fix the absolute luminosity scale. Results today show that CV measurements give $H_0 = 74.03 \pm 1.42 \text{ km Mpc s}^{-1}$. However, a current issue is that at low z , *peculiar motions* tend to contaminate cosmological redshifts, increasing uncertainty in our readings.

4.2.2 Measuring q_0

A requirement for measuring q_0 is the need for a standard candle visible out to $d_L \approx 1000 \text{ Mpc}$ ($z \approx 0.1 \rightarrow 1$). What we use is the Type Ia Supernovae (SNe Ia), which is caused by the explosion of a white dwarf crossing the Chandrasekhar mass limit at $1.4M_\star$. These explosions have a peak brightness of $L \approx 4 \cdot 10^9 L_\star$ (100,000 times more luminous than Cepheids), and are highly consistent to *standardization* (correcting slight variations using light curves).

Recall that for small redshift, we have a luminosity distance

$$d_L \approx \frac{c}{H_0} z \left[1 + \frac{1 - q_0}{2} z \right].$$

The observable we work with is the distance modulus ($m - M$), which has a specific formula

$$m - M = 5 \log_{10} \left(\frac{d_L}{10 \text{ pc}} \right).$$

Then, using our approximation for $d_L(z)$ at low redshift, we have

$$m - M \approx 43.24 - 5 \log_{10} \left(\frac{H_0}{68 \text{ km s}^{-1} \text{ Mpc}^{-1}} \right) + 5 \log_{10}(z) + 1.086(1 - q_0)z.$$

Plotting ($m - M$) over $\log_{10}(z)$, we find that decelerating models predict that the SNe Ia appears *fainter* (thus farther) than expected. Our decelerating models for $q_0 > 0$ predicted that the SNe Ia would appear *brighter* (thus closer). So, our data favours $q_0 < 0$ for *acceleration* of the universe; dark energy (Ω_Λ) tends to dominate the universe.

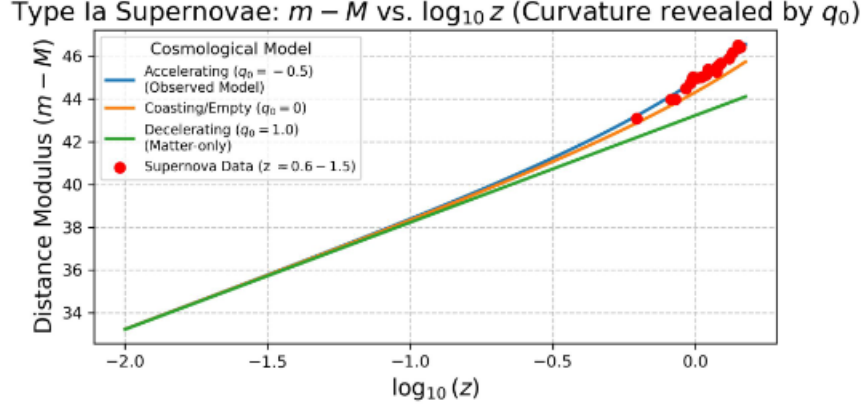


Figure 14: Type Ia Supernovae plot of $(m - M)$ against $\log_{10}(z)$. Curvature is revealed by q_0 .

4.2.3 Independent Confirmation

We can conduct one more independent confirmation through the use of **Baryon Acoustic Oscillations**. The distribution of galaxies provides a standard ruler, called the **BAO Standard Ruler**. Since, in the hot, early universe, sound waves propagated through the photo-baryon fluid, a fixed physical scale called the **sound horizon** became imprinted on the matter distribution at recombination. We observe today that there is a modern co-moving separation of $r = l \approx 150$ Mpc.

We can calculate d_A and q_0 by the observations made by BOSS. We observe an angular separation of the BAO features at high z to be $\delta\theta = 6.438^\circ$ for galaxies at $z = 0.51$. Calculating, we find that

$$d_A = \frac{150 \text{ Mpc}}{(\pi/180) \cdot 6.438^\circ} \approx 1335$$

which is consistent only with an accelerating universe model ($q_0 < 0$). Of course, we naturally conclude that this measurement solidifies the observational evidence for dark energy.

4.3 The Missing Mass Problem

Current cosmological observations (from the CMB and the Large Scale Structure) suggests that the total matter density parameter is

$$\Omega_{m,0} \approx 0.3.$$

This, however, is staggeringly low compared to the total mass density $\Omega_0 = 1$. A central question which arises is: does the mass we can physically account for (stars, gas) match this required total?

To determine the answer, we estimate the mass by measuring light, relying on the **Mass-to-Light Ratio (M/L)**. We filter the spectra of a stellar mass by using a **V-band** ($\sim 500\text{-}590$ nm), which is a standard measure that aligns with the visual peak of the Sun's light output. In fact, the sun's luminosity in the V-band is

$$L_{\star,V} = 4.6 \cdot 10^{25} \text{ W}$$

which is only $\sim 12\%$ of the Sun's total bolometric Luminosity, $L_\star$.

If we consider **light density** (ψ_V) by observing galaxy surveys across the local universe show an average light density of

$$\psi_V = 1.1 \cdot 10^8 L_{\star,V} \text{ Mpc}^{-3}.$$

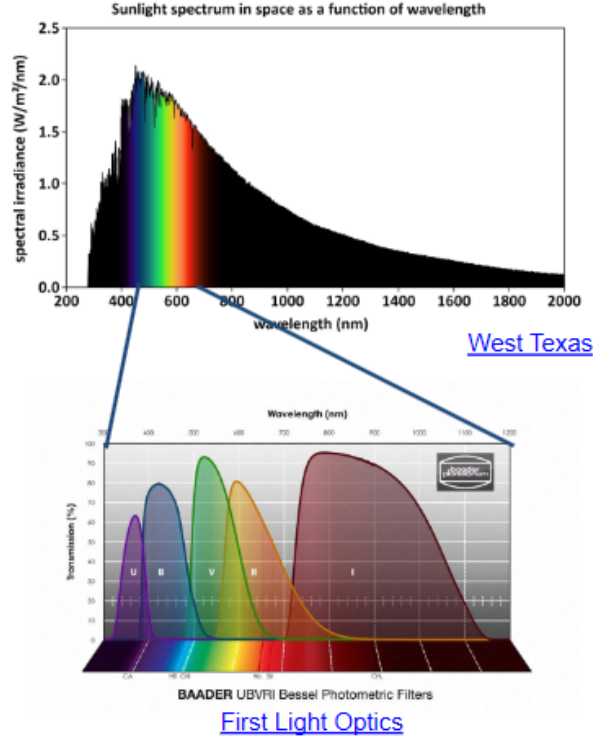


Figure 15: Sunlight spectrum in space as a function of wavelength, decomposed into (zoomed in) its constituent parts.

We see a problem where stellar types vary significantly. **O-type stars** dominate the light, while **M-type stars** (low mass) dominate the total mass inventory. To fix this, we require a **statistical average M/L** ratio to account for the entire population and accurately convert luminosity to mass.

An immediate solution is to define the **Initial Mass Function (IMF)**, which is an empirical distribution of star masses at formation. Massive stars ($M > 1M_{\star}$) follow a steep power law of the form

$$\chi(M > 1M_{\star}) \propto M^{-\beta}$$

suggesting that they are *rare* and burn out quickly. However, low-mass stars ($M < 1M_{\star}$) follow a log-normal distribution, peaking around $0.2M_{\star}$,

$$\chi(M < 1M_{\star}) \propto \frac{1}{M} \exp\left[-\frac{(\ln M - \ln M_c)^2}{2\sigma^2}\right], \quad M_c \approx 0.2, \quad \sigma \approx 0.5.$$

This range is *extremely common and long-lived*.

Further, the IMF appears to be sharply peaked below $\ln(M) = 0$ (**M-stars**). So, we see that M-stars dominate the long-term mass since their lifetimes (billions of years) exceed the short lives of O-stars (~ 3 Myr), skewing the perceived mass balance.

We thus conclude that the stellar mass budget is skewed by the temporary brightness of massive stars.

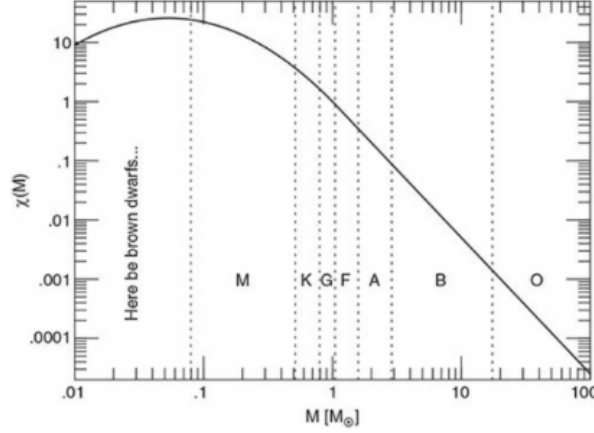


Figure 16: The initial mass function plotted over a range of $M = 1M_{\star}$ to $M = 100M_{\star}$.

If we use the result of the average M/L ratio, we have that it is approximately (and adjusted) $4M_{\star}/L_{\star}$. Further, note that for every O-star, we have ≈ 250 M-stars, with stellar lifetimes of ≈ 3 Myr (O-stars) and $> 10^3$ Myr (M-stars).

The stellar mass density is thus

$$\Omega_{\star,0} = \frac{\rho_{\star,0}}{\rho_{c,0}} \approx \frac{4 \cdot 10^8 M_{\star} \text{ Mpc}^{-3}}{1.28 \cdot 10^{11} M_{\star} \text{ Mpc}^{-3}} \approx 0.003$$

showing that stellar mass accounts only for $\approx 0.1\%$ of the required total $\Omega_{m,0}$. The mass computed is entirely visible and luminous. However, we may broaden this calculation to include stellar remnants, black holes, and brown dwarfs, to show that

$$\Omega_{\star,0} < 0.005.$$

So, we see that the stellar mass cannot account for all the total required matter of $\Omega_{m,0} \approx 0.3$. If we consider the matter due to Baryons, we have also $\Omega_{b,0} \approx 0.048$, which is an absolute limit constrained by the BBN and the CMB (including all protons and neutrons). There is a mass gap of $\approx 80\%$ for the universe, unaccounted for normal matter. Since we are unable to see it, we call this **Dark Matter**.

We can see how dark matter affects galaxies by considering the rotation curves of galaxies. Suppose we wish to measure the orbital speed (v) of stars/gas as a function of radius (R) using Doppler shifts (galactic rotation curves). From Newtonian mechanics, we have that the centrifugal acceleration must balance gravitational acceleration, so

$$\frac{v^2}{R} = \frac{GM(R)}{R^2} \implies v = \sqrt{\frac{GM(R)}{R}}$$

where $M(R)$ is the mass *enclosed* within a radius, R . We expect, since the distribution of visible mass $I(R)$, falls over exponentially by a factor e^{-R/R_s} , that the orbital speed is proportional by

$$v \propto \frac{1}{\sqrt{R}}. \quad (49)$$

However, we observe that spectroscopic measurements show rotational velocity which is constant out to the largest measurable radii. If this is the case, then $M(R)$ must increase linearly with R , so that

$$M(R) = \frac{v^2 R}{G} = 1.05 \cdot 10^{11} M_{\star} \left(\frac{v}{235 \text{ km/s}} \right)^2 \left(\frac{R}{8.2 \text{ kpc}} \right).$$

Thus, the enclosed mass must still grow where luminous matter stops.

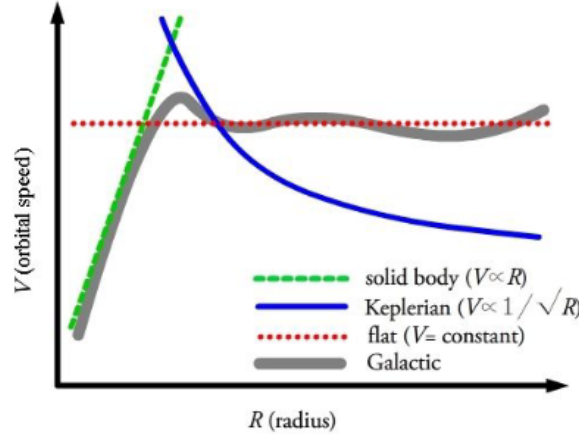


Figure 17: Orbital speed as a function of radius for different curves.

We are forced to conclude then that the missing mass must reside in a giant, roughly spherical, invisible **dark matter halo** surrounding the galaxy. We can compute the required mass-to-light ratio for this halo, which would be

$$\left\langle \frac{M}{L_V} \right\rangle_{\text{gal}} \approx \frac{64 M_{\star}}{L_{\star, V}} \left(\frac{R_{\text{halo}}}{100 \text{ kpc}} \right)$$

which is immensely higher than the stellar ratio

$$\frac{M}{L} \approx \frac{4 M_{\star}}{L_{\star}}.$$

So, we conclude also that dark matter dominates the gravitational potential of individual galaxies. However, this new discovery suggests that high internal galaxy speeds indicate instability, and so should fly apart based on visible mass. Gravity clusters, such as these, must have some other form of stability that maintain the galaxy itself. We can explicitly calculate the minimum total mass needed for this by the **Virial Theorem**.

Since clusters are gravitationally bound and do not expand with Hubble flow, Newtonian physics is fine to use. Let us define the acceleration of any galaxy as a sum of gravitational forces,

$$\ddot{\mathbf{x}} = G \sum_{i \neq j} M_j \frac{\mathbf{x}_j - \mathbf{x}_i}{|\mathbf{x}_j - \mathbf{x}_i|^3}.$$

The total binding energy is defined by the sum of inverse-distance paired masses,

$$W = -\frac{G}{2} \sum_{i, j, i \neq j} \frac{M_i M_j}{|\mathbf{x}_j - \mathbf{x}_i|} = -\alpha \frac{GM^2}{r_h}$$

where r_h is the half-mass radius, and $\alpha \approx 0.45$ is a parameter related to the cluster profile. We determine the kinetic energy by the measured velocity dispersion, which tracks the random kinetic energy of the galaxies,

$$K = \frac{1}{2}M \langle v^2 \rangle, \quad \langle v^2 \rangle = \frac{1}{M} \sum_i M_i |\dot{\mathbf{x}}_i|^2.$$

We set up our derivation by noting that the moment of inertia is of the form

$$I = \sum_i M_i |\mathbf{x}_i|^2$$

so that

$$\ddot{I} = 4K + 2 \sum_i M_i (\mathbf{x}_i \cdot \ddot{\mathbf{x}}_i).$$

Using the binding energy, we have that

$$\sum_i m_i (\mathbf{x}_i \cdot \ddot{\mathbf{x}}_i) = \frac{1}{2} \left[\sum_i m_i (\mathbf{x}_i \cdot \ddot{\mathbf{x}}_i) + \sum_j m_j (\mathbf{x}_j \cdot \ddot{\mathbf{x}}_j) \right] = -\frac{G}{2} \sum_{i,j,i \neq j} \frac{m_i m_j}{|\mathbf{x}_j - \mathbf{x}_i|} = W.$$

Thus, it follows that

$$\ddot{I} = 4K + 2W \tag{50}$$

which is the **Virial Theorem**. In the special case where we have a steady state, we arrive at the **Steady-State Virial Theorem**, which is

$$W = -2K, \quad \ddot{I} = 0.$$

For example, suppose that we are interested in calculating the come mass cluster. For a stable cluster, we have that the steady-state virial theorem holds, so that

$$\frac{1}{2}M \langle v^2 \rangle = \frac{\alpha}{2} \frac{GM^2}{r_h} \implies M = \frac{\langle v^2 \rangle r_h}{\alpha G}.$$

We observe the properties of a redshift of about $z = 0.0232$, with a horizon distance

$$d_{\text{Coma}} = \frac{c \langle z \rangle}{H_0} = 102 \text{ Mpc}.$$

We have, also, that the line of sight (LOS) velocity is $\sigma_r = 880 \text{ km/s}$, with a half-mass radius of $r_h = 1.5 \text{ Mpc}$. The missing cluster mass for the coma cluster follows and is $M_{\text{Coma}} = 2 \cdot 10^{15} M_\star$. Visibly, we can see components making up the cluster to be $\approx 1\%$ of stars and $\approx 10\%$ of hot gas, with a mass-to-light ratio of $M/L \approx 400 M_\star/L_\star$. That is, there is a mass deficit of about 89% and so Dark Matter is required to bind this cluster.

We can confirm our results once more by considering gas dynamics in hydrostatic equilibrium. Suppose we wish to model the hot X-ray gas at $T \approx 10^8 \text{ K}$. Hydrostatic equilibrium shows that

$$\frac{dP_{\text{gas}}}{dr} = -\frac{GM(r)\rho_{\text{gas}}(r)}{r^2}$$

and the ideal gas law,

$$P_{\text{gas}} = \frac{\rho_{\text{gas}}}{\mu} kT.$$

So, we find that this independent, pressure-based method yields a highly consistent total mass estimate,

$$M \approx 1.3 \cdot 10^{15} M_{\star}$$

with a lower limit of $\Omega_{m,0} \approx 0.2$. All of these results show that the three Newtonian methods agree across two scales: the galactic rotation scales, cluster galaxy kinematics, and cluster gas dynamics. That is, the dark matter hypothesis is validated by the mechanics of gravity on all scales.

4.4 Gravitational Lensing

INSERT MISSING NOTES HERE

5 The Cosmic Microwave Background

INSERT MISSING NOTES HERE

At last scattering of the universe, we had a redshift of about $z \approx 1090$. From this, we find that the dark matter, baryon particle, and photonic, density is such that

$$\begin{aligned}\epsilon_{\text{dm}} &= \Omega_{\text{dm}, 0} \epsilon_{\text{c}, 0} (1 + z_{\text{LSS}})^3 \approx 1.7 \cdot 10^{12} \text{ MeV m}^{-3}. \\ \epsilon_{\text{bar}} &= \Omega_{\text{bar}, 0} \epsilon_{\text{c}, 0} (1 + z_{\text{LSS}})^3 \approx 3.1 \cdot 10^{11} \text{ MeV m}^{-3}. \\ \epsilon_{\gamma} &= \Omega_{\gamma, 0} \epsilon_{\text{c}, 0} (1 + z_{\text{LSS}})^4 \approx 3.9 \cdot 10^{11} \text{ MeV m}^{-3}.\end{aligned}$$

We take the ratios relative to baryonic particle densities so that we find

$$\frac{\epsilon_{\text{dm}}}{\epsilon_{\text{bar}}} \approx 5.5, \quad \frac{\epsilon_{\text{bar}}}{\epsilon_{\text{bar}}} = 1, \quad \frac{\epsilon_{\gamma}}{\epsilon_{\text{bar}}} \approx 1.24.$$

To get to us, a given photon must climb out of the potential well formed by dark matter. From Poisson's Equation on the angular diameter, we find that

$$\nabla^2(\delta\phi) = \frac{4\pi G}{c^2} \delta\epsilon$$

where the $\delta\epsilon$ term is driven specifically by dark matter (due to ratios). The full GR calculation (by Sachs & Wolfe in 1967) showed that the Sachs-Wolfe (SW) fluctuations of temperature follow

$$\frac{\delta T}{T} = \frac{1}{3} \frac{\delta\phi}{c^2}. \quad (51)$$

The SW plateau dominates on large scales, telling us the dark matter fluctuations were approximately constant with scale over a wide range of scales.

On small scales, the baryon-photon fluid has pressure from the radiation, inducing oscillations. We have two cases on how the oscillations evolve. If the frequency of the oscillations is small, then gravity will dominate the particles, bringing them to a energetic minimum. On the other hand, if the frequency is larger, then pressure dominates the particles, bringing the particles away from the energetic minimum.

We can link a strong perturbation of the particles with the gravity-dominated case, and a weak perturbation with the pressure-dominated case. That is, we can argue that the oscillations in our CMB power spectrum arise due to these two unique cases. The wavenumber of the oscillations depends on the sound horizon at last scattering. Indeed,

$$d_s(t_{\text{LSS}}) = a(t_{\text{LSS}}) \int_0^{t_{\text{LSS}}} \frac{c_s(t) dt}{a(t)} = \frac{1}{\sqrt{3}} d_{\text{hor}}(t_{\text{LSS}}).$$

The first peak in the CMB power spectrum corresponds to a wavelength of one, so that

$$\theta_s = \frac{d_s(t_{\text{LSS}})}{d_A(t_0)} = \frac{0.145 \text{ Mpc}}{12.8 \text{ Mpc}} \approx 0.7^\circ.$$

From the Planck 2018 Cosmological Parameters, we can specifically calculate the present day baryon and critical densities, which are

$$\begin{aligned}\Omega_{\text{bar}, 0} \left(\frac{H_0}{100} \right)^2 &= 0.02212 \pm 0.00022 \\ \Omega_{\text{c}, 0} \left(\frac{H_0}{100} \right)^2 &= 0.1206 \pm 0.0021.\end{aligned}$$

The peak location is such that $\theta_{\text{MC}} \cdot 100 \approx 1.04077 \pm 0.00047$, with an optical depth of $\tau = 0.0522 \pm 0.0080$.

6 Big Bang Nucleosynthesis (BBN)

We cannot see beyond the last scattering surface, and so we cannot make any direct observations on the universe prior to matter-radiation equality. Recall that the temperature scales as $T \propto a^{-1}$, so that in the radiation dominated portion of the universe, we can write

$$T(9t) \approx 10^{10} \text{ K} \left(\frac{t}{1\text{s}} \right)^{-1/2}, \quad E \approx 2.7kT \approx 3 \text{ MeV} \left(\frac{t}{1\text{s}} \right)^{-1/2}.$$

At BBN, the age of the universe was approximately $t \approx 2 \cdot 10^{-13} \text{ s}$, where the mean particles in the universe had a maximum energy of any particle in the LHC. The BBN covers the first few minutes after the big bang, and deals with processes involving fission and fusion. We know that an atomic nucleus contains Z protons and N neutrons, with the conditions $Z \geq 1$ and $N \geq 0$. We define the mass number as

$$A = Z + N.$$

For example, we may write hydrogen-1 or helium-4 as ${}^1\text{H}$ or ${}^4\text{He}$. Here, 1 and 4 are the mass numbers of these isotopes, and the symbol themselves are the number of protons. That is, we condense the notion further so that we could instead write 11 or 42 . We have a specific exception in the case of **heavy hydrogen**, also known as **deuterium**, which is $D = {}^2\text{H}$.

The binding energy, B_D , is the energy required to pull apart the nucleus into its constituent parts. For example, a proton and neutron nucleus can be decomposed such as

$$p + n \rightleftharpoons D + 2.22 \text{ MeV}, \quad (B_D = 2.22 \text{ MeV}).$$

The binding energy or nucleons tells us the energy released in fission or fusion.

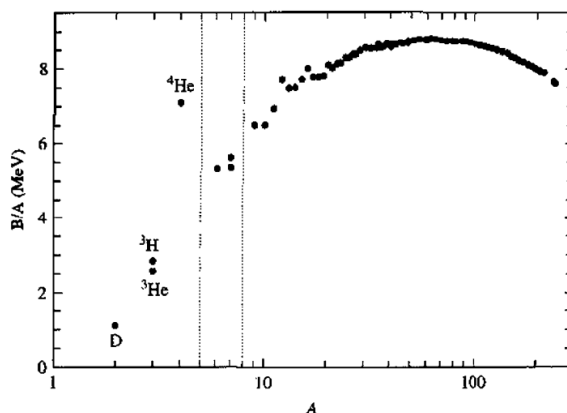


Figure 18: Binding energy per nucleon (B/A) as a function of the number of nucleons in an atomic nucleus. Note the absence of nuclei at $A = 5$ and $A = 8$.

References